

Singapore and Sustainable Development

Goal 1: No Poverty

Measures:

Retirement Adequacy (Central Provident Fund (CPF)): Employers and employees contribute a percentage of employees' monthly gross salaries to employees' CPF accounts. Periodical CPF top-ups to low-income families and individuals. Higher interest rate for first \$60,000 (additional 1%).

Silver Support Scheme caters to bottom 20% of elderly (65 years old and above) through quarterly cash pay-outs (\$300-750).

Employment Assistance (Workfare and Progressive Wage Model): Workfare Income Supplement tops up CPF savings and wages of older low-wage workers in their working years.

Workfare Training Support scheme encourages skill upgrading while Progressive Wage Model helps to increase wages of workers in the cleaning, security, and landscaping sectors through skills upgrading and improved productivity.

Subsidised Public Housing: Government subsidies and ability to pay for down payment and monthly mortgage loan instalments through Ordinary CPF Account.

Affordable Healthcare: Multi-tiered approach to healthcare coverage.

Extensive subsidies in public healthcare institutions, Medisave (compulsory savings scheme for personal or immediate family's hospitalisation, day surgery, and certain outpatient expenses), MediShield Life, ElderShield (insurance schemes designed to help Singaporeans financially through major illnesses or severe disability and offer protection for life).

Patients from lower- and middle-income families receive more subsidies through means-testing.

Pioneer Generation Package (additional 50% off subsidised services and medications at polyclinics and specialist outpatient clinics (even more subsidies for lower- and middle-income Singaporeans), subsidies at participating GP and dental clinics under Community Health Assist Scheme (CHAS), \$1200 a year for those with moderate to severe functional disabilities through Pioneer Generation Disability Assistance Scheme, special premium subsidies for MediShield Life (40-60%) and MediSave top-ups (\$200-800), additional CareShield Life (long-term care insurance scheme that provides basic financial support should Singaporeans become severely disabled, especially during old age, and need personal and medical care for a prolonged duration) participation incentives (additional \$1500)).

Access to Quality Education: Ministry of Education (MOE) Financial Assistance Scheme (for families with \$690 or lower per capita income OR \$2750 or lower gross household income, full subsidisation of school, miscellaneous, textbook, school attire fees, transport subsidy (50% school bus fare, \$120 transport credit per year), \$900 bursary for pre-university students), bursaries and subsidies.

Social Safety Nets:

ComCare: Provides assistance to low-income Singaporeans through four ComCare Schemes: Short-To-Medium-Term Assistance (SMTA), Long-Term Assistance (LTA), ComCare Interim Assistance and Student Care Fee Assistance.

SMTA targets low-income families who temporarily cannot work and need financial support while seeking employment.

LTA targets individuals who are unable to work due to health reasons, have limited or no income and little to no family support. Assistance given includes cash grants, rental and utilities, medical and employment assistance.

Medifund: provides assistance for low-income patients who are unable to pay their medical bills, even after receiving government subsidies and drawing on other means of payments. To provide more targeted assistance for the low-income elderly and the young, Medifund Silver and Medifund Junior were introduced.

Public Rental Scheme: Those who are unable to afford their own homes have the option of applying for public rental flats. Rental costs are heavily subsidised and highly affordable (from S\$26 for a 1-room flat and S\$44 for a 2-room flat). Fresh Start Housing Scheme (ability to buy a short-lease-2-room Flexi flat in Built to Order/Sale of Balance (pool of unsold flats) for 45-65 years with a minimum 20 year occupation period, ability to take up HDB concessionary loan (interest at 0.1% above Ordinary CPF Account Interest rate), cap of \$30,000 on resale levy, grant of \$30,000 (lump sum of \$20,000 right before key collection, remaining \$15,000 over 5 years into Ordinary CPF account equally over 5 years after key collection), priority allocation given in HDB sales exercises of up to 10% of 2-room Flexi units under Tenants' Priority Scheme).

Challenges:

Demographic Changes and Weakening Family Support: Shrinking family sizes, rising divorce rates, and more complex family structures.

Increasing Demands on Social Spending: Demands on social spending expected to mount further as Singapore is one of the fastest-ageing societies in the world. In 2018, one in eight Singaporeans are aged 65 and above. This will increase to one in four by 2030. Resident old-age support ratio is projected to fall by almost half from 5.1 in 2017 to 2.1 in 2030. With a rapidly-ageing population, healthcare expenditure will continue to grow.

Sustaining a Culture of Self-help with the Support and Resources of the Government and Community: Need to avoid inadvertently creating a culture of over-dependence, which runs contrary to Singapore's philosophy of self-reliance. A culture of over-dependence may also lead to abuse of our system and policies, diminishing resources that should be allocated to those who truly need them.

Opportunities:

Continuing to Invest Heavily in Education at All Levels and Lifelong Skills Upgrading: Strengthen the ecosystem of support for children from low-income and vulnerable families through the KidSTART (comprises of KidSTART Home Visitation (regular home visits to parents (or main caregivers) and their infants, from antenatal stage until the child is three years old. Parents receive support in skills and practical knowledge across areas of child growth, health and nutrition. Regular screening of child's development and maternal well-being across the antenatal and postnatal stages to identify and treat issues early. Group-based activities called 'Group Connect' organized for parents to provide peer support and reinforce learning from home visits. Depending on the family's needs, child will be enrolled into preschool programme), KidSTART groups (weekly community-based playgroup sessions, for parents or main caregivers and their toddlers, aged 1 to 3 years old to help parents and caregivers build up skills in child development and parent-child bonding through evidence-based curricula of structured and purposeful play. Parents provided with educational resources to reinforce learning at home. KidSTART Group facilitators may conduct home visits for families who need additional support) and KidSTART Enhanced Support to Preschools (Selected Anchor Operator pre-schools operated by PAP Community Foundation (PCF) and NTUC First Campus are provided with additional resources to improve engagement with and support for parents, and to improve the child's school readiness (addressing barriers related to poor preschool attendance and referrals to other support programmes to ensure their developmental needs are addressed etc)), and invest in the growth of Singaporeans through SkillsFuture.

Ensuring Prudent Social Spending: Introduced several schemes to safeguard individuals against large hospitalisation bills, and subsidies to meet the growing demand for caregiving services.

Stepped up support for low-income and vulnerable seniors.

A Community Network for Seniors rolled out to strengthen coordination and collaboration between government agencies and community-based stakeholders to leverage each other's strengths and resources as they reach out, befriend, and support seniors living in the community.

More Coordinated and Effective Social Service Delivery: Strengthen collaboration and coordination across SSOs, government agencies, and community partners in order to provide better and more targeted assistance.

Goal 2: Zero Hunger

Measures:

Import Source Diversification: Adoption of Science-based risk assessment approach to ensure safety of food imports. Singapore imports food from over 170 countries.

Optimising Local Food Production: Farms encouraged to explore innovations in food productions, such as indoor multi-tier farming automation and precision agriculture via sensors and Internet of Things (IOT).

“Farm Transformation Map” to spur transformation and worked with local producers to optimise use of space and improve productivity to reduce reliance on manpower and better mitigate environmental risks.

Established Agriculture Productivity Fund (comprises of Basic Capability Upgrading (BCU) Scheme (supports the purchase of equipment that would help increase farm’s productivity such as increase in production, manpower savings and resource savings), Productivity Enhancement (PE) Scheme (supports the adoption of automated, advanced and integrated farming systems through either Standard (supports farming systems through capability upgrading, increased production capacity and land intensification) or Test-bedding (encourages farms to develop, test and demonstrate feasibility of new farming technologies/systems that are customised and optimised to Singapore’s conditions or for farm’s own needs) and Cash Advancement (APF projects with approved funding of \$30,000 and above may be eligible for up to 30% cash advancement)).

Agri-Food and Veterinary Authority (AVA) provides technical assistance and training to farmers, and conducts R&D with farms and in partnership with tertiary and research institutes.

Internalisation: Opens new markets and helps farms overcome land constraints in Singapore by encouraging local farms and food companies to venture overseas.

Also encourage those companies to re-export some of the produced food to Singapore for local consumption.

Stockpiling: Rice Stockpile Scheme (RSS) makes it mandatory for rice importers to hold a two-month stockpile in government warehouses to ensure there is an adequate supply of rice in the market and to stabilise prices in times of short-term shortages.

Multilateral food stock arrangements to enhance Singapore's food resilience.

Singapore is part of the ASEAN Plus Three Emergency Rice Reserve (APTERR), which aims to ensure the availability and accessibility of rice during a regional food emergency.

Reducing Food Wastage: Partnered with stakeholders across food supply chain through holistic food waste management strategy.

Publicity programme "Love Your Food-Waste Less. Save More." to encourage consumers to make prudent choices in food purchase, preparation and storage habits

Collaborated with local community organisations, grassroots leaders and stakeholders on ground-up initiatives to reduce food waste.

For students, "Love Your Food @ Schools" Project hands-on way to learn about the closed-loop process in food waste reduction and recycling.

AVA and National Environment Agency (NEA) with various industry stakeholders developed food waste minimisation guidebooks for food retail establishments, supermarkets, and food manufacturing establishments.

Improvement of post-harvest management and storage to prolong shelf-lives.

Assisting the Vulnerable: ComCare programme to assist low-income households to meet basic needs. Includes a monthly cash quantum for food expenses. Immediate assistance can be provided on the spot.

Various Voluntary Welfare Organisations (VWO) deliver meals to low-income families.

Challenges:

Impact of Climate Change: Rapid urbanisation and industrialisation on a global scale exacerbated the impact of climate change, threatening the farming and food production sectors. Changes in the frequency and severity of droughts and floods could pose challenges for farmers in sustaining their crop yields. Warmer waters could also disrupt ecosystems, threatening fishery supplies.

Spikes in Food Consumption: Singapore's growing population and middle class will increase demand and food consumption, putting additional stress on our food supplies.

Opportunities:

Innovation and Technology to Maximise Food Production: Developing a progressive farm sector will help to attract a new generation of agri-specialists to further regional food security efforts.

New Sources and Changing Mindsets: Need to continue diversifying import sources.

Tapping VWOs' knowledge on preventing food wastage and educating our young on prudent food consumption practices.

Sino-Singapore Jilin Food Zone (SSJFZ) in Jilin, China: First commercially-driven agriculture project between Singapore and China. Focuses on developing an integrated value chain from agribusiness R&D, animal and pasture input, production, processing and other supporting industries, logistics, and sale. Upon completion of first phase development in 2021, the integrated farm will comprise 33 farms, an abattoir, and a high-end processing plant across 150 hectares. The farm is expected to produce 300,000 pigs annually, with 100,000 meant for Singapore.

Vertical Farming: Sky Greens is the world's first low carbon, hydraulic-driven vertical farm. Developed a patented vertical farming system which comprises multi-layer

troughs in a rotating A-shape aluminium vertical frame. The frame can be as high as 9 metres with 38 tiers of growing troughs.

Goal 3: Good Health and Well-Being

Measures:

Accessible and Improved healthcare: Healthy life expectancy at birth was 73.6 years in 2016, and the highest in the world.

Screening for breast and cervical cancers at all governmental clinics is heavily subsidised.

Women's Health Advisory Committee set up to equip women with the knowledge and skills to look after their own health.

Agency for Integrated Care (AIC) helps seniors access eldercare services and works with the eldercare sector to expand and improve services.

Several funds available to support the elderly, such as Seniors' Mobility and Enabling Fund (subsidises assistive devices and home care consumables).

AIC conducts outreach to promote active ageing, and encourages community partners to befriend and care for vulnerable seniors.

Control of Communicable Diseases: National Childhood Immunisation Schedule (NCIS) and National Adult Immunisation Schedule (NAIS) recommends vaccinations for children and adults.

Singapore-Field Epidemiology Training Programme set up within the Ministry of Health (MOH) to develop capability in conducting epidemiological surveillance, investigation, and response to outbreaks and other public health emergencies.

Healthcare Epidemiology Team established within MOH to strengthen the national response to outbreaks of hospital-acquired infections, including Multi Drug Resistant Organisms (MDROs).

National Outbreak Response Team formed to rapidly mobilise public health experts to facilitate the national outbreak response.

Singapore continuously engages the international community at international and regional platforms such as WHO and ASEAN to fight the global spread of disease.

Control of Non-Communicable Diseases: Several health campaigns have been introduced to encourage positive changes in Singaporeans' lifestyles, such as increased physical activity and healthier eating choices.

Adopted a multi-pronged approach towards tobacco control, comprising strategies such as public education, provision of smoking cessation services, legislation controlling tobacco advertising and sales of cigarettes to minors, and taxation.

Challenges:

Increasing Healthcare Expenditure due to an Ageing Population: Rapidly ageing population means healthcare expenditures to rise.

Opportunities:

Innovative Medical and Health Technology and Solutions: Growing number of healthcare start-ups in areas such as chronic disease management, digital diagnostics, and consumer health and wellness.

Singaporeans relatively receptive to healthcare technology (4 in 10 own a health application/gadget).

Goal 4: Quality Education

Measures:

Quality, Affordable, and Accessible Education for All: Compulsory Education Act mandates all Singaporean Citizen children above 6 years old must be enrolled in national primary schools up to Primary Six.

School fees are heavily subsidised and bursary schemes are available for those who need additional support.

Singapore students emerged top in all three categories (reading, mathematics, and science) and in the Collaborative Problem Solving portion of the Programme for International Student Assessment (PISA).

Early Childhood Development: Establishment of MOE Kindergartens and developing the Nurturing Early Learners Curriculum (provides a comprehensive range of resources targeted at 4 to 6 years old and acts as a pedagogical guide for preschool educators to understand how children learn and to design learning activities to cater to the children's unique needs, abilities, and learning styles).

Post-Secondary Education: Six Autonomous Universities (AUs), five polytechnics, and the Institute of Technical Education (ITE) offer programmes that play to individuals' different strengths.

Universities offer a broad spectrum of undergraduate and postgraduate degrees, some in partnership with top foreign universities such as the Yale-National University of Singapore (NUS) College, the first liberal arts college in Singapore.

ITE provides students with technical knowledge and vocational training.

Polytechnics impart industry-relevant skills through diploma courses.

Maximising the Potential of Students with Special Needs: 75% of children with reported SEN attend mainstream schools.

Specialised curricula have been introduced in the 20 Government-funded Special Education (SPED) schools, which are run by VWOs in close consultation with MOE.

Curriculum Framework for SPED schools, “Living, Learning, Working in the 21st Century” introduced to provide SPED schools with a common direction for designing and delivering a quality holistic education to meet the needs of these students.

Framework for Vocational Education in SPED schools was launched to establish a structured vocational education programme that enhances students’ employability and boosts their confidence to lead independent lives.

Established partnerships between SPED schools and mainstream schools which aim to provide social and academic integration opportunities.

Schools have barrier-free accessibility, with handicapped facilities provided at ground or entry level. All newly-built schools and those undergoing major upgrading have barrier-free enhancements such as lifts, ramps, and handicapped toilets, in compliance with the prevailing Barrier Free Accessibility Code.

Lifelong Learning: SkillsFuture is a national movement that aims to promote industry support for individuals to acquire the skills necessary to progress in their career as well as foster a culture of lifelong learning.

Strengthening of partnerships between education institutions and industries to co-develop more learning options that are industry-relevant and provide exposure to real work environments.

Education for Sustainable Development: Character and Citizenship Education (CCE) aims to inculcate values in our children so that they become civic-minded and responsible individuals and citizens.

International Outreach: Schools, universities, and the private sector offer scholarships to international students to create diverse student communities to foster better cross-

cultural understanding and global awareness among local students and facilitates mutual understanding and goodwill.

Challenges:

An Increasingly Uncertain Economic Landscape: Millions of people will be joining the global marketplace, and technology is driving rapid change, making it challenging to remain relevant in an increasingly uncertain landscape.

Widening Social Gap: Widening income gap and growing inequality in society poses an obstacle in ensuring that our educational system continues to allow for high social mobility.

Over-emphasis on Academic Grades: Risk of over-emphasis on academic grades, at the expense of non-academic interests.

Opportunities:

Equipping Students with 21st Century Competencies and Cultivating a Culture of Lifelong Learning: 21st Century Competencies Framework focuses on ensuring that the next generation continues to be highly-skilled, flexible, and adaptable.

Mindset shift among employees and employers to continuously acquire and hone their skills to enhance their work-readiness is central to the SkillsFuture effort.

Goal 5: Gender Equality

Measures:

End Discrimination Against Women and Girls: Singapore part of United Nations Convention on the Elimination of All Forms of Discrimination against Women (CEDAW).

Inter-Ministry Committee (IMC) on CEDAW created to implement and monitor policies and initiatives to address women's needs, in accordance with CEDAW. IMC supported by the Office for Women's Development (OWD) in the Ministry of Social and Family Development (MSF). OWD also national focal point for women matters in Singapore.

In October 2017, Singapore presented Fifth Periodic Report to the UN Committee on the Elimination of Discrimination against Women.

Eliminate Violence Against Women and Girls: Violence against women addressed through legislative framework, a multi-stakeholder approach, training and professional competency, and public education.

Government agencies work closely with NGOs and the community to foster violence-free environment.

MSF launched a "Break the Silence Against Family Violence" campaign in 2016.

Inter-Agency Taskforce on Trafficking-in-Persons (TIP) formed to coordinate decisions and align TIP policies between agencies.

Singapore's anti-TIP law, the Prevention of Human Trafficking Act (PHTA) (criminalises TIP in the form of sex, labour and organ trafficking), came into force in March 2015.

Ratified the UN Protocol to Prevent, Suppress and Punish Trafficking in Persons.

Promote Shared Responsibility Within the Household and the Family: Worked closely with the Centre for Fathering, in catalysing the "Dads for Life" movement to encourage active fathering.

Introduced leave provisions for fathers, which have been enhanced over the years (in 2017, paid Paternity Leave extended to two weeks, and increased Shared Parental Leave to four weeks (from one week previously)). Child care and infant care leave provisions extended equally to working fathers and mothers.

Introduced a new Tripartite Standard (a set of actionable and verifiable set of practices agreed upon by the tripartite partners, which are voluntary for employers to adopt. Employers which adopt this commit to providing up to four weeks of unpaid leave for employees with pre-term/multiple births or babies with congenital conditions, and up to two weeks of unpaid leave for employees with immediate family members who are hospitalised). In March 2018, to encourage employers to provide additional unpaid leave for employees with dependents who have unexpected care needs (e.g. pre-term births), regardless of their employees' gender.

Tripartite Committee on Work-Life Strategy (comprises members from government agencies, tripartite partners and employers, promotes work-life practices as a strategy to optimise business performance and facilitate employees to manage their own work-life fit) launched an Advisory on the implementation of flexible work arrangements. Tripartite partners also launched a Tripartite Standard on Flexible Work Arrangements in 2017 to further improve adoption of it.

Women in Leadership: Women's Register was established by the Singapore Council of Women's Organisations (SCWO), to assist women who would like to volunteer their services to the community, make new connections in social and professional arenas, and obtain support and guidance from women leaders.

BoardAgender (initiative of SCWO launched in 2011 with the support of MSF) facilitates greater awareness of benefits of gender-balanced business, and encourages and enables women to contribute their expertise in the boardroom and through committees.

Diversity Task Force regarding Women on Boards (DTF) released recommendations to ensure there is gender diversity on boards and senior management.

Diversity Action Committee, comprising illustrious business leaders and professionals from the private, people and public sectors, formed to drive improvements to address the issue of underrepresentation of women on boards.

Women make up 45% of the volunteers in grassroots organisations under the People's Association (network of 1,800 grassroots organisations).

In the Singapore Government, there were currently seven women political office-holders in 2018 (three women ministers in the Cabinet, two mayors chairing the Community Development Councils (oversee various districts) and Madam Halimah Yacob (Speaker of Parliament from January 2013 until September 2017 when she became first woman President of Singapore)

Women's Health: Women's and Children's hospital dedicated to the advancement of women's health. Hospital has evolved to become a regional leader in obstetrics, gynaecology, paediatrics and neonatology.

State of the World's Mothers Report 2015 ranked Singapore 14th out of 179 countries and the best place in Asia to be a mother.

Life expectancy at birth for women increased from 82.6 years in 2006 to 85.1 years in 2016.

Only two registered maternal deaths in 2016.

Infant mortality rate low at 2.4 per thousand resident live births in 2016.

Mortality rates for children under five years old dropped from 2.9 per thousand resident live births in 2006 to 2.7 in 2016.

Women's Access to Economic Resources: Muslim women in Singapore can receive better protection on matters of inheritance (under Muslim inheritance law, men are by default, apportioned a greater share of inheritance than women. However, MUIS Fatwa Committee issues fatwas (formal ruling or interpretation on a point of Islamic law given by a qualified legal scholar (known as a mufti)) from time to time to protect the financial welfare of Muslim women and their dependents.

Administration of Muslim Law Act (AMLA) amended in 2017 to allow the court to permit women beneficiaries to become sole administrators of a deceased man's estate.

Use of Enabling Technology: 30% of 199,800-strong info-communications and media workforce is female.

Since the launch of TechSkills Accelerator Programme (TESA), about 33% of participants in Info-Communications and Media Development authority (IMDA)'s human capital development programmes with placement outcomes is female.

29% of local university and polytechnic students who took part in IMDA's internship and mentorship programmes are female.

Policies and Legislation for the Promotion Of Gender Equality And The Empowerment Of Women And Girls: Women's Health Advisory Committee set up in 2012 to promote the health and well-being of women. The Committee was revamped in 2016 as the Women's Health Committee and focuses on key health issues among women in Singapore (increasing cancer screening uptake, promoting bone health, and fighting diabetes, including gestational diabetes).

Specific legislation in place to protect the rights of women (Women's Charter, Penal Code, Children and Young Persons Act, and the Protection from Harassment Act etc).

Challenges:

Low Representation on Boards: Women only held 10.8% of board seats in companies listed on the Singapore Stock Exchange as of December 2017.

Low Workforce Participation by Women due to Work-Life Imbalances and Challenges: Besides careers, women tend to adopt the role of caregiver for their children or elderly family members, in addition to undertaking household chores. As a result, they may forego their careers to prioritise their obligations at home

Potential Retirement Inadequacy of Women: With longer life spans and shorter careers as a result of family responsibilities, women are more likely than men to be financially vulnerable when they reach old age.

Opportunities:

Building up Women's Representation through the Diversity Action Committee: e Diversity Action Committee (DAC) was formed in 2014 to promote women's representation on the boards of companies in Singapore and expand the pool of board-ready women. In April 2017, DAC set a triple-tier target of 20% by 2020, 25% by 2025 and 30% by 2030 for women's representation on boards of Singapore Stock Exchange-listed companies.

Support in Old Age: Initiatives such as Retirement and Re-employment Act (retirement age between 62 to 67 years old and can be re-employed) enable seniors who are willing and able to continue working beyond retirement to do so in order to boost their retirement savings.

Silver Support Scheme and Pioneer Generation Package enhance financial assistance for older or retired women.

Goal 6: Clean Water and Sanitation

Measures:

Safe and Affordable Drinking Water: Ranked 170th out of 190 countries in terms of freshwater availability

“Four National Taps” strategy: Water from local catchments, imported water from Malaysia, Water reuse – the NEWater Initiative and Seawater desalination.

Expanded local catchment areas to two-thirds of Singapore’s land area.

NEWater produced using advanced membrane technologies and ultraviolet disinfection to further purify treated water. Highly valued by the industry and primarily supplied to non-domestic sectors such as wafer fabrication parks, industrial estates and commercial buildings for industrial and cooling purposes.

Improvements in membrane technology significantly reduced the cost of desalination. Singapore has three desalination plants with a combined total capacity of 130 million imperial gallons per day.

NEWater and desalinated water are expected to meet up to 85% of Singapore’s water needs by 2060.

Public Utilities Board (PUB) monitors and assesses our water, in line with over 300 different water quality parameters, exceeding the requirements stipulated under international drinking water regulations.

Fish Activity Monitoring System (FAMS) (uses video analytics technology to monitor if there is fish kill which could be an indication of possible toxicity in the water, and sends alerts to the operator when this phenomenon occurs) and other online monitoring systems and lab tests, which enhance the security measure of drinking water by providing constant monitoring of water quality.

Sanitation and Hygiene: 100% of wastewater collected and treated.

In 2025, with the completion of the Deep Tunnel Sewerage System (DTSS), Singapore’s long-term needs for wastewater collection, treatment, reclamation and disposal will be met in a cost-effective and sustainable way. Completed DTSS will also

streamline and further close the water loop through NEWater by increasing the percentage of recycled water in Singapore's water supply.

Efficient Water Use: Provided funding to companies to implement water efficiency initiatives, such as consultations for solutions, and water stewardship.

Focus on industrial water-saving solutions through improving rates of recycling and increasing uptake of water conservation projects.

Number of programmes in place to assist the private sector achieve higher water efficiencies (e.g. Mandatory Water Efficiency Management Plan (mandate for large water users with net water consumption of at least 60,000 m³ to submit a notification to PUB for their different sites meeting this water consumption threshold, install private water meters at key water usage areas within their premises, submit their annual Water Efficiency Management Plan (WEMP) to PUB and appoint at least one Water Efficiency Manager), Water Efficiency Awards (for premises with lowest Water Efficiency Index (WEI) or industrial plants with highest recycling rates) and Water Efficiency Fund (can provide a grant of 50% of the cost of water efficiency assessment (subject to a cap of \$30,000), a grant of 50% of the cost of the pilot study (subject to a cap of \$50,000, 50% given on commission of pilot plant, other 50% given on approval of report submitted to PUB), \$0.71 for every m³ of PW/\$0.45 for every m³ of NW/\$0.37 for every m³ of IW saved over the economic life of the water recycling plant or 15 years (whichever is earlier) OR 50% of capital cost of water recycling plant (both subject to a cap of \$1 million per project and 50% given on commission of water recycling plant and other 50% given on third year after commissioning of the water recycling plant with reasonable water savings achieved and the approval of the report (must have summary of the proposed recycling project and technology, including the potential water savings and actual water savings achieved for the first 3 years on a monthly basis and challenges faced during implementation/operation of project) submitted to PUB), \$0.71 for every m³ of PW / \$0.45 for every m³ of NW / \$0.37 for every m³ of IW saved over the economic life of the equipment or 15 years (whichever is earlier) OR 50% of the Total Qualifying Cost (50% given on completion of installation of equipment and other 50% given after one year of monitoring water savings and approval of the report submitted to PUB (report must have summary of proposed monitoring method

and evaluation of water savings achieved over 1 year, challenges faced during operation of equipment and potential areas to improve/optimize water savings achievable)).

Mandatory measures for households and industries to use water efficiently such as maximum allowable flow rates for taps and mixers, and maximum allowable flushing capacities for flushing cisterns and urinal flush valves.

Under the Water Efficiency Labelling Scheme, water efficiency labels are also required for water fittings and appliances.

Engaging Communities on the Value of Water and Sanitation: Clean-up of Singapore River which involved the relocation of thousands of street hawkers, squatters and polluting industries, and the removal of over 250 tonnes of rubbish that had accumulated in the river and on its banks through dredging the river bank and bed, allowing aquatic life to return and thrive and laid a foundation for a reservoir.

Mouth of the Marina Channel in the southern part of Singapore dammed through the construction of the Marina Barrage in 2008, resulting in a steady water supply, flood control, and a recreational space for the community.

Launch of the Active, Beautiful, Clean Waters (ABC Waters) Programme in 2006 to transform utilitarian drains and canals into attractive waterways, improving runoff quality using green cleansing features. 36 ABC Waters site completed as of January 2018.

Friends of Waters programme recognises the efforts of the community in spreading water-related messages and keeping Singapore's water supply sustainable.

Worked with schools through MOE to include water conservation topics in the national curriculum.

Worked with partners from the "3P sectors" (People, Public and Private) to commemorate Singapore World Water Day annually through month-long celebrations in March, which coincide with the UN's World Water Day on 22 March.

International Cooperation: Water and sanitation success the result of having benefitted from technical assistance and cooperation with other countries and international organisations such as the World Bank and UN agencies during the early years of nationhood.

Established the Singapore Water Academy, a practitioner-focused learning institute in urban water management. In 2016, the Singapore Water Academy worked with the Ministry of Foreign Affairs to conduct water-related programmes for ASEAN countries.

Engaged international organisations regularly on water issues, such as UNESCO, the International Water Association, the International Desalination Association, the World Bank, the Asian Development Bank and the Asia Pacific Water Forum, and contribute actively to the discourse on sustainable urban water management.

Organises the biennial Singapore International Water Week (SIWW), a premier global platform to share and co-create innovative water solutions.

Lee Kuan Yew Water Prize, an international water prize, honours outstanding contributions by individuals or organisations in solving the world's water challenges. For example, the laureate for 2018, Professor Rita Colwell, was conferred the Water Prize for her pioneering insights into microbial water quality surveillance, which has significantly contributed to the understanding and prevention of waterborne diseases.

Challenges:

Water Security and Weather Resilience: Local catchments and imported water dependent on rainfall. Dry weather can threaten the reliability of our water supply from these sources.

Increased Water Demand: Water use is expected to increase from 55% of water demand in 2018, to 70% by 2060.

Opportunities:

Cutting-Edge Technology: Invested in weather-resilient water sources (NEWater and desalinated water).

Small amount of NEWater is injected into local reservoirs to maintain healthy stock levels.

Water from the reservoirs is treated for potable use through a series of water treatment processes at the water treatment plants.

By 2020, Singapore will have two additional 30 mgd desalination plants.

Expecting desalinated water and NEWater to meet up to 85% of our water needs by 2060.

Plans to scale up electrodeionisation (EDI) technology, which uses an electric field to extract dissolved salts from water to reduce energy usage of desalinisation.

\$670 million in funding set aside to promote R&D and grow water industry.

Infrastructure Planning and Innovation: Plans to enhance wastewater sewer network through construction of the DTSS (comprises a network of linked sewers leading to two major tunnels. These deep tunnel sewers convey wastewater by gravity to centralised water reclamation plants located at our coastal areas. Enhances the reliability of our wastewater system by minimising the risk of cross contamination between water catchments and wastewater. By 2025, Phase 2 of the DTSS will extend the existing system to western Singapore with a 40-kilometre long deep tunnel linked to 60 kilometres of sewers. When fully completed, DTSS will shrink the land occupied by wastewater infrastructure in Singapore by 50%).

Worked with industry partners to develop a Smart Water Grid, a network of wireless sensors installed in potable water supply mains across Singapore, which functions as a real-time platform to monitor water pressure, flow and quality. System provides decision support tools for network management and allows early detection of anomalous network occurrences, enhancing the efficiency of water supply to consumers. This helps minimise losses of water due to leaks and ensures a reliable water supply.

Marina Barrage: 350-metre wide dam across the mouth of the Marina Channel in southern Singapore.

Serves a catchment area of one-sixth the size of Singapore.

Desalted Marina Reservoir through a natural replacement process.

During heavy rain, a series of nine crest gates at the dam is activated to release excess stormwater into the sea when the tide is low. During high tide, giant pumps can drain excess stormwater into the sea.

Water level in Marina Reservoir kept constant as it is unaffected by tides, making it ideal for water-based recreational activity. Green roof of Marina Barrage popular recreation spot.

Goal 7: Affordable and Clean Energy

Measures:

Access to Electricity and Energy Services: Between 1963 and 1969, 300 electrification schemes were implemented, and the Government provided S\$1 million monthly to subsidise the electrification of areas furthest away from the existing power grid.

Mobile Collection Unit was set up in 1964.

Rural electrification programme, which cost nearly S\$20 million, enabled more than 200,000 people to enjoy the benefits of electrification.

Power grids in rural areas consisted of overhead bare and stranded copper conductors supported on poles. The electricity supply was therefore frequently disrupted, by weather-related events (e.g. storms or tree fall). It was also easy to pilfer electricity or even steal the wiring itself. This issue was resolved when the overhead lines were replaced with insulated wires in the 1980s.

Implemented a compulsory electrification scheme when new public housing units were built by HDB.

Residential and commercial consumers have access to services providing town gas and liquefied petroleum gas (LPG) for cooking and water heating.

Reliability of The Electrical System: Singapore instituted a vocational training programme in schools to build up the technical competency of our local workforce.

By 1984, there were about 670,000 electricity consumers, with an annual demand of 13,000 million kilowatt-hours (kWh).

Computerised electricity system in the 1980s with the introduction of the Power System Control Centre to monitor and control the operation of Singapore's power stations and transmission network.

Condition Monitoring Systems were also introduced progressively to provide early warning of any impending failure of transmission and distribution equipment.

Computerised maps of electricity cables and gas pipelines to avoid accidental damage to underground cables and disruptions in supply to consumers during construction of high-rise public housing and the public transport railway system.

Energy Market Authority (EMA) put in place regulatory measures to further strengthen the reliability of Singapore's electricity supply.

EMA conducts regular reviews with industry players to enhance the emergency preparedness of the power sector.

Average electricity interruption time of less than one minute per customer per year.

Security of Gas Supply: 95% of our electricity is generated using natural gas, which is imported via licensed term and spot gas importers. Most of Singapore's natural gas is imported from Indonesia and Malaysia through pipelines.

Since May 2013, Singapore has also started importing liquefied natural gas (LNG) to diversify energy sources and strengthen energy security. Singapore LNG Terminal was built in May 2013 with two storage tanks and an initial throughput capacity of 3.5 million tonnes per annum (Mtpa). As of September 2017, the terminal's throughput has been increased to around 11 Mtpa, with the completion of additional regasification facilities. Further expansion of the terminal is in progress and Singapore will see the addition of a fourth tank in 2018.

Competitive Electricity Market: First country in Asia to liberalise electricity market.

Allows the market to set the price of electricity, without any subsidies.

Opened up the electricity generation and retail markets to commercial players, established a regulatory framework and introduced a wholesale electricity market with spot bidding every 30 minutes.

Consumers with a monthly consumption of at least 2,000 kWh are contestable, and can buy electricity from retailers. Non-contestable consumers, mainly households and small businesses, buy electricity at the regulated tariff.

Retail market liberalised for the rest of Singapore from the second half of 2018.

Sustainable Energy: Part of the Paris Agreement on climate change

Competition in the liberalised power generation sector has spurred power generation companies to switch from steam plants powered with fuel oil to more efficient Combined Cycle Gas Turbines (CCGTs) fuelled by natural gas, the cleanest fossil fuel.

Solar photovoltaic (PV) deployment in Singapore has grown rapidly, with total installed capacity increasing from 0.4 megawatt peak (MWp) in 2008 to around 143 MWp in 2017. Plans to further raise solar adoption to 350 MWp by 2020 and further to one gigawatt peak (GWp) beyond 2020.

Investment in research, development and demonstration (RD&D) of solar PV and related energy technologies, addressing market barriers to deployment, enhancing regulatory framework for intermittent generation sources, and having the public sector take the lead in adopting solar energy to encourage growth of the solar industry.

International Cooperation: Committed to being a constructive and active player in discussions on sustainable energy at multilateral fora such as ASEAN, APEC (Asia-Pacific Economic Cooperation) and G20 (group of countries consisting of Argentina, Australia, Brazil, Canada, China, European Union, France, Germany, India, Indonesia, Italy, Japan, Mexico, Russia, Saudi Arabia, South Africa, South Korea, Turkey, United Kingdom and the United States), as well as through our engagement with international organisations such as the International Energy Agency (IEA) and International Renewable Energy Agency (IRENA).

Under ASEAN Chairmanship in 2018, Singapore hosted the 36th ASEAN Ministers on Energy Meeting (AMEM) as chairman of the ASEAN energy track.

Singapore organises the annual Singapore International Energy Week (SIEW) to discuss global energy issues.

Joined the IEA as an Association Country in 2016. Partnered with the IEA on two initiatives: the Singapore-IEA Regional Training Hub and Singapore-IEA Forum.

Worked with the IEA to host the inaugural IEA Energy Efficiency Training Week in Singapore in July 2017 under the first initiative, and the Singapore-IEA Forum at SIEW in October 2017.

In 2018, Singapore partnered with the IEA to host a training programme on energy investments and a workshop on digitalisation for ASEAN and to hold the second Singapore-IEA Forum which focused on digitalisation.

Challenges:

Energy Security and Diversification of Energy Sources: Limited renewable energy options, with most electricity generated using natural gas. Lack of hydro resources, low wind speeds and mean tidal range and geothermal energy not economically viable.

Reducing Singapore's Emission Intensity: Under Paris Agreement, Singapore pledged to reduce greenhouse intensity by 36% from 2005 levels by 2030.

Conducive Regulatory Environment for Power generation Investments: In anticipation of an increasing share of renewables, need to ensure conducive regulatory environment that facilitates power generation investments

Opportunities: LNG enhances energy security as it allows us to source for natural gas globally. Launched a two-stage Request for Proposal (RFP) in June 2014 to appoint up to two new LNG importers to supply Singapore's next tranche of LNG. LNG importers to supply Singapore's next tranche of LNG. Appointment of two new LNG importers and the flexibility for any gas user to source LNG from the spot market contribute to the development of more competitive and dynamic gas and electricity markets in Singapore.

Investment in system-level solutions such as solar forecasting and energy storage technologies to manage intermittency.

Two consortiums were appointed to implement Singapore's first utility-scale Energy Storage System (ESS). Project aims to evaluate the performance of different ESS technologies under Singapore's hot, humid and highly urbanised operating environment to help establish clear technical guidelines for ESS deployment (e.g. grid connection and safety requirements for installation).

EMA launched consultation paper to seek industry feedback on policy framework for energy storage systems.

Deploying Mitigation Measures: Enhanced the Energy Conservation Act to strengthen energy efficiency practices among companies.

Plans to implement Minimum Energy Performance Standards (MEPS) for common industrial equipment.

Provides grants and support to help companies perform energy audits, enhance energy efficiency and reduce emissions.

Implementation of carbon tax across all sectors without exemption.

Infrastructure Planning and Innovation: Singapore Electricity Market Outlook (SEMO) launched to provide information and visibility on the longer term outlook of the energy landscape in Singapore, such as the projected electricity system demand and supply conditions. Worked with research institutes to feature the outlook of solar PV generation in Singapore. This will help the industry better understand the characteristics of solar PVs in Singapore and facilitate investment decisions for such technologies.

Floating Solar Test-Beds: One MWp floating solar PV system test-bed was launched at Tengeh Reservoir comprising rows of floating solar panels atop a hectare of water.

Far performed better than rooftop solar panels because of cooler temperatures in the surrounding environment.

Initiative a joint collaboration between Economic Development Board, the Solar Energy Institute of Singapore under the National University of Singapore, and PUB.

Possibility of large-scale floating solar PV system (up to 50 MWp) at Tengeh Reservoir. Energy generated could power about 12,500 four-room flats.

Goal 8: Decent Work and Infrastructure

Measures:

Restructuring Singapore's Economy:

Overall real productivity grew by 2.5% p.a. between 2009 and 2016. Real median wage grew by 2.6% p.a. over the same period.

Set up the Committee on the Future Economy (CFE) to adopt a forward-looking posture in order to develop economic strategies to position Singapore well for the future. Addresses areas crucial to Singapore's future economic development: future growth industries and markets, corporate capabilities and innovation, jobs and skills, urban development and infrastructure and connectivity.

Transforming the Local Workforce: Ministry of Manpower introduced the "Adapt and Grow" initiative to help Singaporeans affected by economic slowdown and restructuring adapt and stay relevant in the labour market

Various programmes under "Adapt and Grow" provide wage and training support to both employers and jobseekers. For example, workers including those who leave their jobs mid-career can "re-skill" to enter new job roles and growth industries such as logistics, financial services and pharmaceuticals through the Professional Conversion Programme (PCP). Others can take part in Work-Trials which place them on attachments to companies for on-the-job training. More than 24,000 individuals were able to find jobs through the "Adapt and Grow" initiative in 2017. Out of these, over 50% were Professionals, Managers, Executives and Technicians (PMETs) and about 30% had experienced long-term unemployment of six months or more.

Challenges:

Impact of global shifts on Singapore's small and open economy: Structural shifts in external environment and demographic trends: Global growth has been subdued and is expected to be lower than in the previous decade.

Global productivity growth has been sluggish.

Global value chains are changing, with major trading nations including China looking to in-source more.

Populations are ageing.

Rapid technological change: Innovation cycles have shortened.

New technologies can supplant entire industries and displace workers such as robotics and artificial intelligence programs displacing routine jobs in manufacturing and services.

Shift in mood away from globalisation: Nativist politics and protectionist economics growing in strength and influence in Europe and the United States.

Anti-globalisation trend will undermine international trade, hurting small open economies like Singapore, with two-thirds of GDP generated by external demand.

Opportunities:

Leverage opportunities to innovate, deepen capabilities, remain connected and stay relevant: CFE identified seven mutually reinforcing strategies:

Deepen and diversify our international connections

Acquire and utilise deep skills

Strengthen enterprise capabilities to innovate and scale up

Build strong digital capabilities

Develop a vibrant and connected city of opportunity

Develop and implement Industry Transformation Maps

Partner each other to enable innovation and growth

Sustainable Tourism: Singapore Tourism Board driving sustainable practices in the tourism sector

Developing workforce capacity and capability: STB developed a Skills Framework (SF) in partnership with employers, training providers, industry associations and unions.

The SF provides up-to-date information on employment, career pathways, job roles, existing and emerging skills and competencies, as well as relevant education and training programmes. Individuals can then use this to make informed choices on career development and skills upgrading, while employers can better recognise skills and invest in skills training for their employees.

Training Industry Professional in Tourism (TIP-iT) incentive scheme supports companies in employee upgrading as well as talent and leadership development.

Building local engagement and partnerships: STB works closely with the precinct champions (e.g. the Orchard Road Business Association which works with stakeholders on Singapore's main shopping belt), content providers as well as other public agencies and private stakeholders to profile the precincts through differentiated events.

STB works closely with the Chinatown Business Association (CBA) and local community to organise festive celebrations, form business collaborations, and address issues for the betterment of the surrounding citizens and businesses to improve visitor experience and inject vibrancy to the precincts and cultural enclaves

When revamping Chinatown Food Street, STB and Chinatown Business Association (CBA) turned to Singaporeans to get their opinions on how the Food Street could be reimagined.

In Little India, STB worked closely with the Little India Shopkeepers and Heritage Association to organise place-making initiatives such as the Deepavali Light Up and the ARTWALK Little India event, a public art collaboration between LASALLE College of the Arts and STB.

Offers funding support via its Association Development Fund to tourism-related precinct associations for capability building and to help these associations become self-sustaining in the long term.

Goal 9: Industry, Innovation and Infrastructure

Measures:

Infrastructure Building: Singapore is one of the world's busiest container ports, which handles more than 33 million Twenty-foot Equivalent Units of containers and is linked to over 600 ports globally. One of the world's best airports, which serves more than 62 million passengers per annum and is connected to more than 400 cities.

Next generation mega container port being constructed at Tuas, which will double our current container handling capacity.

Started planning for airport expansion at Changi East, which will add a fifth airport terminal and additional infrastructure to serve another 50 million passengers per year by around 2030.

Built an extensive public transport network, with a well-connected rail network forming the backbone. By 2030, the rail network will expand from 230 kilometres today to 360 kilometres, and eight in 10 households will be within a ten-minute walk of a train station.

As part of the Infocomm Media 2025 Plan, Singapore aims to improve access to information technologies and improve our citizens' quality of life.

Recently introduced a fourth telecommunications provider (TPG Telecom) through a New Entrant Spectrum Auction as well as allowing mobile virtual network operators to provide services. These widen options for consumers, ensure a competitive market, and keep telecommunications and internet services affordable and innovative.

As of November 2017, Singapore has a mobile phone penetration rate of 149.6%.

As of November 2017, the Wireless Broadband Population Penetration Rate is 206.1%.

Wireless@SG programme provides nationwide free Wi-Fi hotspot services, which offers internet access to users in thousands of public locations like public libraries, train stations and shopping malls.

Sustainable Industrialisation supported by Innovation: Established the Agency for Science, Technology and Research (A*STAR), which consists of 23 Research

Institutes and Centres, to develop innovative technology to further economic growth and to improve lives.

Developed the one-north district as an R&D hub focused on catalysing research in biomedical sciences and engineering. Today, one-north is equipped with state-of-the-art facilities for Multinational Corporations (MNCs) to undertake innovation-intensive activities in Singapore, facilitate technology transfer to our local enterprises, and has become the cradle to Singapore's vibrant start-up ecosystem. Biopolis and Fusionopolis, the main developments within one-north, home to over 250 companies, 600 start-ups and 16,000 scientists, researchers and innovators from both public and private sectors.

Singapore consistently ranks among the top ten countries under the Global Innovative Index, published by Cornell University, INSEAD and the World Intellectual Property Organisation.

In 2017, Singapore ranked first in the Innovation Input Sub-Index and second in the Infrastructure sub-pillar.

Continued investment in R&D is also evident in creation of more than 50,000 jobs in the R&D sector in 2016, at a Compound Annual Growth Rate of 3% over the previous ten years.

Challenges:

Rapid Technological Change: New technologies can displace entire industries, making workers redundant, even as they create new opportunities.

Helping SMEs stay competitive: Most research-intensive and innovative small economies in the world have large home-grown companies that are MNCs but Singapore's domestic enterprises are not as large or research-intensive.

Opportunities:

Committee on the Future Economy Strategies: CFE had identified strategies allow our population to cope with the resulting uncertainties and grasp new opportunities:

Acquire and utilise deep skills

Strengthen enterprise capabilities to innovate and scale up

Build strong digital capabilities

Partner each other to enable innovation and growth

Strengthen Enterprise Capabilities to Innovate and Scale-up: Government can help our enterprises stay competitive through the following strategies:

Strengthen our innovation ecosystem: Companies can tap on the RIE2020 Plan to develop innovative and viable commercial products and establish commercially-oriented entities to commercialise research findings and intellectual property

Support enterprises to scale up: Provide targeted assistance to high-growth enterprises to scale up and internationalise and encourage partnerships between large and small enterprises

Catalyse the private sector to provide more growth capital: Simplify the venture capital regulatory framework and attract more Private Equity firms to Singapore

Singapore's Future of Manufacturing initiative: Aims to drive the adoption of advanced manufacturing technologies among enterprises through public-private partnerships.

Model Factories allows the ecosystem of research performers, end users, technology providers and system integrators to jointly innovate, test and demonstrate FoM technologies. A*STAR launched the Model Factory@SIMTech in October 2017, and the Model Factory@Advanced Remanufacturing and Technology Centre (ARTC) in 2018 to allow companies, particularly SMEs, to learn about the latest manufacturing technologies and collaborate with stakeholders to test-bed and jointly develop innovative solutions. End goal is for SMEs to adopt, in whole or in part, the Model Factory platform technologies on their own factory floors.

Goal 10: Reduced Inequality

Measures:

Reducing Inequalities and Ensuring Social Mobility: Home ownership rate of 90.9% as of 2016.

Ethnic Integration Policy for public housing helped Singapore avoid large ethnic concentrations in particular neighbourhoods.

Workfare Income Supplement (WIS) scheme supplements the wages of older low-wage workers in their working years and tops up their CPF savings. For workers in sectors that may require more help, Progressive Wage Model (PWM) helps to increase wages of workers through upgrading skills and improving productivity.

From 2012 to 2017, income growth at the 20th percentile of full-time employed residents was 4.2% per annum in real terms. Income growth at the median was 3.4% per annum in real terms. Between 2012 and 2017, income growth per household member ranged between 4.2% and 4.6% for each of the lowest five deciles, while the average income growth per household member was 3.5% for the population.

14.3% of Singaporean children from households in the lowest 20% income bracket managed to progress to the top 20% income bracket.

Significant investment in the early childhood sector, and piloting a new early intervention programme for children in low-income and vulnerable families called KidSTART.

Expansion of school-based Student Care Centres (SCCs) and places to cater to the educational, social, and emotional well-being of students, especially those from disadvantaged families.

Robust Social Security System: Singapore's social security system comprises the four pillars of home ownership, healthcare assurance, retirement adequacy through the CPF system, and WIS and Silver Support schemes.

Lower- to middle-income Singaporeans are eligible for higher subsidies for subsidised treatment at public hospital specialist outpatient clinics (SOCs), and higher medication

subsidies at subsidised SOCs and polyclinics. Can also apply for the Community Health Assist Scheme (CHAS), and receive subsidies for treatment at participating general practitioner (GP) and dental clinics.

Introduced structural premium subsidies for lower- to middle-income Singaporeans, as well as Additional Premium Support for those who are unable to afford their premiums even after subsidies.

ComCare provides financial assistance to low-income individuals and families to meet their basic needs, while working hand-in-hand with the families to improve their situation and regain self-reliance.

Policies such as Workfare (supplementing income and retirement savings through cash payments and CPF contributions) designed to encourage people to stay in the workforce and to keep building on their skills.

Empowering and Promoting the Social, Economic, and Political Inclusion of All: Prime Minister Mr Lee Kuan Yew said that Singapore is “not a Malay nation, not a Chinese nation, not an Indian nation. Everybody will have his place: equal; language, culture, religion.”

Established Group Representation Constituencies (GRCs) which are larger electoral divisions represented by between three and six Members of Parliament (MPs), of which at least one MP must belong to a minority ethnic group.

Introduced the Nominated Member of Parliament (NMP) scheme to promote political inclusion of all Singaporeans, including those who may be potentially disadvantaged and marginalised. They represent the varied interests of different groups of Singaporeans. For example, NMP Ms Chia Yong Yong, who is also the President of the SPD, has spoken in Parliament to champion issues concerning the welfare of persons with disabilities.

Introduced Third Enabling Masterplan (2017 - 2021), which looks at improving the quality of life of persons with disabilities, supporting their caregivers, and building a community that is more caring and inclusive. Endorsed by a Steering Committee which

included key stakeholders such as persons with disabilities, caregivers, professionals in the disability field, voluntary welfare organisations, and Government agencies.

WorkPro scheme to help employers implement flexible work arrangements and age-friendly workplaces, to help women and older workers remain in the workforce for as long as they want to.

Challenges:

Globalisation and Technological Disruption: Mitigating the effects of globalisation and technological disruption will continue to be a challenge for any global city. Maintaining social mobility also becomes harder with each successive generation. This is compounded by changing demographics as Singapore's society ages and family sizes shrink.

Opportunities:

Lifelong Learning and Risk-pooling: Introduced more social risk-pooling through CPF LIFE and MediShield Life for retirement and healthcare needs respectively

KidSTART: Comprises of KidSTART Home Visitation (regular home visits to parents (or main caregivers) and their infants, from antenatal stage until the child is three years old. Parents receive support in skills and practical knowledge across areas of child growth, health and nutrition. Regular screening of child's development and maternal well-being across the antenatal and postnatal stages to identify and treat issues early. Group-based activities called 'Group Connect' organized for parents to provide peer support and reinforce learning from home visits. Depending on the family's needs, child will be enrolled into preschool programme), KidSTART groups (weekly community-based playgroup sessions, for parents or main caregivers and their toddlers, aged 1 to 3 years old to help parents and caregivers build up skills in child development and parent-child bonding through evidence-based curricula of structured and purposeful play. Parents provided with educational resources to reinforce learning at home. KidSTART Group facilitators may conduct home visits for families who need

additional support) and KidSTART Enhanced Support to Preschools (Selected Anchor Operator pre-schools operated by PAP Community Foundation (PCF) and NTUC First Campus are provided with additional resources to improve engagement with and support for parents, and to improve the child's school readiness (addressing barriers related to poor preschool attendance and referrals to other support programmes to ensure their developmental needs are addressed etc).

Goal 11: Sustainable Cities and Communities

Measures:

Integrated Land Use Planning: Urban Redevelopment Authority (URA)'s Concept Plan, a long-term strategic land use and transportation plan that outlines broad strategies to guide development for the next 40 to 50 years. Broad long-term strategies are translated into the more detailed Master Plan, which lays out upcoming plans and developments in the next ten to 15 years. Concept Plan is reviewed at least once every ten years and the Master Plan once every five years to take factors such as changing land use needs, socio-economic and technological trends, demographic changes, the economic, social, and environmental needs of current and future generations, as well as regional and global economic developments into account.

In Master Plan 2014, the Jurong Lake District in the western part of Singapore was identified as a new mixed-use business district.

New approaches to district-planning include integrating utilities, facilities and services at the district level instead of at a building level, such as district cooling systems and common services tunnels to achieve economies of scale and cost savings, and enjoy the convenience of tapping on shared services.

Hawker centres co-located with community amenities and offer a variety of high quality and affordable food.

Housing a Nation: Over 80% of Singapore's resident population live in public housing built by the Housing and Development Board (HDB). More than nine in ten of these households in public housing own their flats.

Public housing is heavily subsidised, with a progressive system of housing grants, on top of subsidised purchase prices for new HDB flats. As a result, most first-time home buyers today use less than a quarter of their monthly income to pay their housing loans, well below the international benchmark of 30% to 35% of monthly income allocated to housing.

Most HDB towns are developed based on the “Neighbourhood Principle”, where several neighbourhoods are grouped around a town centre that provides essential services within close reach of the residents.

Punggol, HDB’s youngest town in the northeast of Singapore, is based on a newer planning concept where smaller residential estates share a common green and a variety of well-integrated facilities to enhance accessibility and encourage clean commuting.

Building Safety: Building and Construction Authority (BCA) champions a strong culture of safety awareness and regulation in the built environment sector.

Design and construction of buildings in Singapore are regulated under the Building Control Act and Regulations. Includes a rigorous system of checks and controls throughout the entire building lifecycle of design, construction, commissioning the building before occupation, and maintenance after completion.

Under the Periodic Structural Inspection regime, regular inspections must be conducted on completed buildings by professional engineers to inspect and assess the building condition and recommend rectification measures.

BCA’s regulatory control extends to lifts and escalators. Owners must obtain a permit from BCA for each lift and escalator, carry out monthly maintenance, and test them annually. Contractors must maintain the lifts and escalators in accordance with manufacturers’ recommendations and relevant standards.

Transit-oriented Development and Planning: Land transport strategies and measures are guided by the Land Transport Master Plan, which is reviewed every five years.

Examples of Singapore’s transit-oriented initiatives:

Promoting Public Transport: Aim of 75% of morning and evening peak journeys to be made using public transport by 2030, and at least 85% by 2050.

Singapore’s rail network will be expanded from 230 kilometres today to 360 kilometres by 2030, enabling eight in ten households to be within a ten-minute walk of a train

station, and 85% of public transport journeys of less than 20 kilometres to be completed within 60 minutes.

In 2012, the Bus Service Enhancement Programme was introduced to provide commuters with better connectivity, more comfortable journeys, and shorter waiting times.

Between 2012 and 2017, 1,000 Government-funded buses added and rolled out 80 new bus services to improve connectivity to major transport nodes and key community and commercial facilities.

Walking and Cycling Plan: “Walk Cycle Ride SG” aims to make walking, cycling, and riding public transport a way of life for Singaporeans.

Introduced a Walking and Cycling Plan (WCP) for developments with high pedestrian and cyclist traffic which requires developers to ensure that their designs meet the needs of pedestrians and cyclists, instead of catering mainly to vehicular traffic. It also provides for the building of ramps for barrier-free access.

Developers are incentivised to provide bicycle lots and supporting facilities through exemption of these spaces from the gross floor area calculation.

120 kilometres of sheltered walkways have been constructed, with 200 kilometres planned for completion by end-2018.

National Cycling Plan (NCP) envisions cycling as an integral part of Singapore’s transport system. Revised in 2013 with more ambitious targets: provide every HDB town with a cycling path network, and build an island-wide off-road cycling path network of over 700 kilometres by 2030.

Inclusive Transport: All train stations have been equipped with at least one barrier-free entrance with a lift, a tactile guidance system and wheelchair-accessible toilets.

More than 85% of train stations have barrier-free access routes from the station entrance to the station platforms.

Priority queue zones for passengers in need for boarding of trains, public buses and lifts.

Public buses are also equipped to allow parents to board with children in open strollers and are wheelchair-accessible.

Safer Streets: 50 “Silver Zones” will be implemented by 2023 in areas with high senior resident populations, amenities which seniors frequent, or higher accident rates involving senior pedestrians. Silver Zones have road safety features, such as lower speed limits, centre dividers, and road humps and chicanes that slow down motorists and remind them to look out for pedestrians.

Green Buildings: Singapore is working to ensure that at least 80% of our total building gross floor area will be green by 2030.

Launched the BCA Green Mark Scheme in January 2005 to promote resource efficiency and reduce any potential environmental impact in the built environment. Paved the way for the formulation of Singapore’s first Green Building Masterplan in 2006, which focused on new buildings. Masterplan was revised twice in 2009 and 2014 to cover existing buildings and tenanted space, and drive the built environment sector to meet the 80% green buildings target.

Singapore has more than 3,200 green buildings with a total gross floor area of more than 94 million square metres, equivalent to more than 34% of total floor area of all buildings in Singapore.

Green Transport: Introduced the Fuel Economy Labelling Scheme which provides information on the fuel efficiency of each vehicle model.

introduced a scheme that provides rebates for low-emission vehicles and levies surcharges for high-emission ones.

Rolled out an electric car-sharing programme, BlueSG, which will introduce 1,000 shared electric cars and 2,000 charging kiosks island-wide by 2020.

Deployed 60 electric buses in mid-2020 for trials.

Green Spaces: Close to 10% of land in Singapore is set aside for parks and nature conservation.

More than 80% of households live within 400 metres or a ten-minute walk to a park. Aim to expand this to more than 90% of households by 2030, by creating more neighbourhood and regional parks.

One of the most iconic regional parks is the Singapore Botanic Gardens (SBG) which is Singapore's first United Nations Educational, Scientific and Cultural Organization (UNESCO) World Heritage Site.

National Parks Board (NParks) has identified six key areas to fulfil this vision: Establish world-class gardens

Rejuvenate urban parks and enliven our streetscape

Optimise urban spaces for greenery and recreation

Enrich biodiversity in our urban environment

Enhance competencies of our landscape and horticultural industry

Engage and inspire communities to co-create a greener Singapore

BCA and United Nations (UN) Environment Collaborations: Centre for Sustainable Buildings, established through the signing of the 2nd BCA-UNEP Memorandum of Understanding (MOU) in 2011, provides technical support, tools, and solutions for the building sector in Asia.

In 2013, the BCA and UN Environment established a five-year partnership project, the "Nationally Appropriate Mitigation Action (NAMA) Development for the Building Sector in Asia", which supports four participating Asian countries: Indonesia, the Philippines, Thailand, and Vietnam, in developing national plans to reduce greenhouse gas emissions in their building sectors. Project is part of the International Climate Initiative supported by the German Federal Ministry for the Environment, Nature Conservation and Nuclear Safety.

Global Alliance for Buildings and Construction (Global ABC) Regional Roundtable for Asia-Pacific: Initiative launched at the 21st Conference of the Parties to the UN Framework Convention on Climate Change (UNFCCC), as part of the Lima-Paris Action Agenda. Aims to scale up actions within the buildings and construction sector to reduce emissions.

Hosted the Global ABC's inaugural Regional Roundtable for Asia Pacific in conjunction with the International Green Building Conference in September 2016. The two-day event involved a series of activities to meet the global sustainable buildings and climate change agenda. Included a presentation of the UN Environment's Finance Initiative publication "Sustainable Real Estate Investment – Implementing the Paris Climate Agreement: An Action Framework", and the Working Session of the Sustainable Energy for All (SE4All) Building Efficiency Accelerator. Attended by more than 85 participants from national governments, local authorities, NGOs, international financial institutions and research institutions from 14 countries in the region.

Centre for Liveable Cities' International Capacity Development Programmes: Since 2015, Centre for Liveable Cities (CLC) has organised international capacity development programmes for 1,472 international city leaders, officials, and practitioners from 42 countries.

CLC's international programmes are guided by the Singapore Liveability Framework, which identifies the outcomes and systems of a liveable and sustainable city. The Framework is also a means to assess sustainable urban development as outlined in SDG 11 and the New Urban Agenda (action-oriented document that mobilizes Member States of the UN and other key stakeholders to drive sustainable urban development at the local level).

In October 2016, Singapore signed a three-year MOU with the UN Human Settlements Programmes (UN-Habitat) to jointly develop international capacity development programmes.

SG UN-Habitat International Leaders in Urban Governance Programme was held from 5 to 9 June 2017. More than 40 participants from 14 cities representing 12 African

countries attended the programme which shared Singapore's strategies in urban transformation contextualised to African cities' needs.

URA Academy: Conducts training sessions to share Singapore's planning experience and expertise with overseas government officials and professionals. Each year, it hosts more than 2,500 delegates from over 45 countries.

Organises a four-day Integrated Land Use Planning course since 2009 to provide a comprehensive overview of URA's core work.

Challenges:

Limited Land: Need to continually optimise the use of our limited land, whether through redevelopment, planning, or building underground in order to keep up with demand.

Changing Demographics: Population expected to age rapidly, with the number of Singaporeans over 65 years of age and above doubling to 900,000 by 2030. Dependency ratio also expected to decrease, with only two working adults supporting each elderly person by 2030.

Increasing Diverse Population: "One-size-fits-all" approach to urban planning will no longer be able to cater to a more diverse resident mix.

Ageing Infrastructure: Need to ensure that buildings are safe and well-maintained to sustain economic and population growth.

Opportunities:

New growth Districts: Redevelopment of Greater Southern Waterfront region after relocation of existing maritime ports to the western part of Singapore, and redevelopment of Paya Lebar region in the northeast, after relocation of the existing military air base.

Spatial Strategies: Setting up economic centres outside the traditional business and financial district in the central region, such as Changi Business Park in the East and one-north in the West.

Planning for more residential spaces in central Singapore to enable more people to work nearer their homes.

Underground Infrastructure: Some examples under study include an underground goods mover system to reduce freight transport on roads, underground electrical substations and rock caverns for storm water drainage and storage to increase our water resilience.

An Enabling City: Introduced barrier-free accessibility, more seating and community spaces in housing estates.

Programme to equip flats with senior-friendly fittings such as grab bars and slip-resistant floor tiles.

Integrated spaces with senior facilities such as day care, and health and community programmes in estates

Enhanced parks with senior-friendly amenities (piloting a network of ten therapeutic pocket gardens based on horticulture therapy to support seniors with dementia and post-stroke patients through the provision of contemplative spaces and activity zones etc).

Help seniors live close to their family and community with priority schemes for new HDB flats, a Proximity Housing Grant (\$30,000 for families, \$15,000 for individuals living together with parents. \$20,000 for families and \$10,000 for individuals living near (within 4 km) parents) for resale flats, and purpose-built housing options such as three-generation “3-Gen” flats and short lease two-room “Flexi” flats.

Exploration of integrating senior-friendly housing and senior care services, such as assisted living developments.

Varied Housing Options: Public housing environment offers a range of options to meet varying needs and demands, such as different budgets, designs, and locations to cater to home buyers with different aspirations and income levels, as well as changing family structures.

Design for Maintainability: Encouraged consultants and developers to consider maintainability outcomes further upstream (design buildings that are easier to maintain).

Worked with industry stakeholders to develop the Design for Maintainability Checklist in 2016 and a Façade Access Design Guide in 2017 to motivate designers and developers to integrate suitable solutions at the design stage for safer, more labour-efficient and cost-effective maintenance regimes.

Maintenance of Exterior Features: Building owners are required to ensure that any exterior features (e.g. windows, claddings, plaster) of their buildings are maintained and securely fixed.

Introduced a Periodic Façade Inspection regime, which focuses on buildings older than 20 years. Inspections will be required once every seven years to allow for early detection of façade deterioration for timely repair.

Sino-Singapore Tianjin Eco-City: Singapore and China embarked on a flagship bilateral project to jointly develop the Sino-Singapore Tianjin Eco-city (SSTEC). The 30 square kilometre SSTEC provides a platform for both countries to collaborate on environmental protection, resources and energy conservation, and build a harmonious society.

Prior to the development of the SSTEC, the site comprised mainly uninhabitable salt pans, barren saline land and polluted water bodies.

As of May 2018, the eight square kilometre Start-Up Area is largely completed. The SSTEC is now home to about 80,000 residents and about 5,000 registered companies.

SSTEC's development is guided by a Key Performance Indicator (KPI) Framework. The Framework comprises 26 KPIs spanning environmental, economic and social aspects, including 100% green buildings, 100% water potability at tap, more than 20% renewable energy usage, 100% services network coverage, and 100% provision of free recreational and sports facilities within 500 metres of residential areas.

Actively pursuit in eco-developments such as rehabilitating its 2.6 square kilometre wastewater pond, conducting research in green development, developing green building standards and constructing green buildings, tapping renewable energy sources, encouraging residents to lead environmentally-conscious lifestyles, and promoting green transport.

Singapore and China working together to document lessons and experiences gained in the last ten years to share with other cities in China and beyond.

World Cities Summit: Platform for government leaders and industry experts to address liveable and sustainable city challenges, share integrated urban solutions, and forge new partnerships.

Jointly organised by CLC and URA.

Includes the Lee Kuan Yew World City Prize Lecture (honours outstanding contributions by individuals or organisations towards solving the world's water challenges), the World Cities Summit Mayors Forum (discuss pressing urban challenges and share the best practices), and the Young Leaders Symposium (select group of change-makers from diverse sectors to come together to discuss common challenges, exchange best practices and also establish stronger networks amongst city leaders around the world).

Organised during the Urban Sustainability Week in Singapore, during which the Singapore International Water Week (SIWW) (connect the water industry for innovative solutions and sustainable urban water management) and the CleanEnviro Summit (CESS) (global networking platform for thought leaders, senior government officials, regulators and policy makers, and industry captains) are also held.

6th WCS in 2018, under the theme of “Liveable & Sustainable Cities: Embracing the Future through Innovation and Collaboration” explored how cities can be made more liveable and resilient through better governance and planning, technology and social innovations, as well as collaborations with stakeholders and other cities.

In line with Singapore’s 2018 ASEAN Chairmanship, a number of ASEAN focused events were also organised, including the 6th Meeting of Governors and Mayors of ASEAN Capitals on 7 July 2018.

Goal 12: Responsible Consumption and Production

Measures:

Waste Reduction: Zero Waste Nation philosophy is integrated into national plans and policies, such as the Sustainable Singapore Blueprint.

In 2017, Singapore generated 7.7 million tonnes of waste.

Working to increase our national recycling rate from the current 61% to 70% by 2030.

Encouraging the 3Rs (Reduce, Reuse, Recycle): Singapore generates approximately 60,000 tonnes of e-waste each year.

E-waste recycling is conducted through a public-private partnership that forms a network of collection centres across Singapore where consumers can voluntarily and conveniently deposit their e-waste for recycling.

Introducing a mandatory e-waste management framework based on the principle of Extended Producer Responsibility (EPR) in 2021.

Mandate that businesses report on the type and amount of packaging they put on the market and their plans for reduction by 2021.

Step up engagement with stakeholders and businesses to cut down the excessive use of plastic bags and disposables like single-use food containers.

Created an integrated waste management system to collect all municipal waste to prevent litter. They are then sent directly for incineration at waste-to-energy plants, which are equipped with air pollution control equipment to meet emissions limits for pollutants such as sulphur dioxide, dioxins and furans.

Households and the industry each generate about half the amount of food that is wasted.

Encouraged food manufacturers, retail food establishments and supermarkets to re-price or re-distribute unsold or excess food to consumers or donate them to charities.

Encourage consumers to reduce food waste through publicity and outreach initiatives that encourage smart and prudent food purchases, preparation and storage habits.

By supporting on-site food waste treatment at commercial premises and hawker centres, as well as piloting district-level food waste treatment, food waste recycling rate increased from 12% in 2012 to 16% in 2017.

Recycling bin is provided for every public housing block and landed housing unit. Private residential developments also required to provide recycling receptacles within the estate.

Since 2014, all new public housing projects are fitted with a dual-chute system on every floor for the separate collection of recyclables and general waste. This is also mandatory for new private non-landed residential developments taller than four storeys since April 2018.

Community Involvement: NEA collaborates with schools to set up recycling corners.

Preschool 3R Awareness Kit, consisting of a set of picture cards and a Teacher's Guide, assists kindergarten teachers in planning activities to interest pre-schoolers in practising the 3Rs and educating them on what and how to recycle.

Community 3R Outreach Programme (CROP) seeks to raise awareness of the 3Rs through public outreach initiatives. All 3R community events and initiatives carry a common tagline: "Reduce, Reuse, Recycle. Care for Our Environment."

Launched a video for households available on the Clean Green Singapore YouTube channel.

Included a 3R module in the myENV mobile app, for members of the public to learn more about the 3Rs and locate the nearest recycling collection points, including e-waste collection points and Cash-for-Trash locations. The app also features pop-up messages on the positive impact of recycling which serve as regular reminders to everyday consumers.

Corporate Engagement: Developed various online 3R guidebooks for businesses to encourage implementation of 3R practices.

3R Awards for Hotels and Shopping Malls recognise outstanding 3R efforts to minimise and recycle waste.

In 2014, Singapore mandated the reporting of waste data and waste reduction plans by large commercial premises.

Singapore Packaging Agreement (SPA) launched in 2007 as a joint initiative by the Government, industry and NGOs. SPA signatories encouraged to redesign their products and processes to enjoy cost savings. This helps reduce packaging waste from consumer products and in supply chain. The SPA also offers a platform where companies share experiences, exchange practical ideas and collaborate on cost-effective solutions to reduce waste, including through a packaging benchmarking database. Over the past decade, SPA signatories have cumulatively reduced about 39,000 tonnes of packaging waste and saved more than S\$93 million in material costs.

Under the SPA, an eco-label, the Logo for Products with Reduced Packaging (LPRP), introduced to enable consumers to identify products with reduced packaging.

Industry Development: Promote growth of recycling industry and creation of higher value-added jobs.

Investment in R&D to develop solutions to extract value and resources from key waste and residue waste streams.

Launched the Environmental Services Industry Transformation Map to improve productivity, promote growth and create better jobs for the cleaning and waste management sectors in Singapore.

Sound management of Hazardous Chemicals and Toxic Industrial Waste: Implemented stringent regulatory frameworks to ensure the environmentally sound management of hazardous chemicals and toxic industrial wastes throughout their life cycle, in accordance with international multilateral environmental agreements (MEAs).

Party to several MEAs that provide guidelines on regulatory and implementation guidance, including in areas with transboundary implications: the Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and Their Disposal, Stockholm Convention on Persistent Organic Pollutants, Rotterdam Convention on the Prior Informed Consent Procedure for Certain Hazardous Chemicals and Pesticides in International Trade, the Vienna Convention for the Protection of the Ozone Layer and the Montreal Protocol on Substances that Deplete the Ozone Layer, and most recently, the Minamata Convention on Mercury.

Control of Hazardous Substances and Products Containing Controlled Hazardous Substances: Controls the import, export, transport, sale, storage and use of hazardous substances, under the Environmental Protection and Management Act (EPMA) and the Environmental Protection and Management (Hazardous Substances) Regulations (EPM (HS) Regulations).

Any individual planning to import, export, transport, sell, store or use hazardous substances locally must first obtain a HS licence or permit. Approval also required to transport hazardous substances in quantities exceeding the limits in the EPM(HS) Regulations.

Put in place requirements such as periodic inspections for bulk packaging, maximum allowable transportation quantities, approved transportation route and timing, as well as the submission of transportation emergency response plans to ensure the safe transportation of hazardous substances.

Conducts inspections to audit the records of hazardous substances maintained by entities with HS licences and permits.

Electronically process inward and outward declarations for the import and export of hazardous substances through the Whole-of-Government (WOG) TradeNet computerised network system.

Regulates the import, export and local use of products containing hazardous substances controlled under the EPMA, through the same regulatory framework and systems described above. This includes, for instance, non-electronic measuring devices such as barometers and thermometers, as well as high pressure mercury vapour lamps for general lighting purposes which are to be phased out by 2020 under the Minamata Convention (international treaty designed to protect human health and the environment from anthropogenic emissions and releases of mercury and mercury compounds).

Implemented the Restriction of Hazardous Substances ("SG-RoHS") framework which came into effect on 1 June 2017 to restrict the amount of hazardous substances such as lead, chromium and cadmium entering the environment from Electrical and Electronic Equipment (EEE).

Control of Toxic Industrial Waste: Environmental Public Health (Toxic Industrial Waste) Regulations require all toxic industrial waste collectors to be licensed to carry out treatment, reprocessing and disposal of toxic industrial wastes. Approval also required to transport toxic industrial wastes exceeding the quantities stipulated in the Regulations.

NEA implemented the Hazardous Waste (Control of Export, Import and Transit) Act to ensure Singapore meets its obligations as a Party to the Basel Convention (international treaty that was designed to reduce the movements of hazardous waste between nations, and specifically to prevent transfer of hazardous waste from developed to less developed countries (LDCs)) on the Control of Transboundary Movement of Hazardous Wastes and Their Disposal. This includes the implementation of the Prior-Informed Consent (PIC) Procedure (mechanism for formally obtaining and disseminating the decisions of importing Parties as to whether they wish to receive future shipments of those chemicals listed in Annex III (list of hazardous chemicals) of the Convention and for ensuring compliance with these decisions by exporting Parties) under the framework of the Convention.

Encouraging Energy-Efficient and Water-Efficient Behaviour and Practices:

Mandatory Energy Labelling Scheme: Introduced the Mandatory Energy Labelling Scheme (MELS) for household air-conditioners and refrigerators to help consumers compare their energy efficiency and make more informed purchasing decisions. Scheme was extended to clothes dryers, televisions and incandescent lamps and their direct replacement. In 2014, design of the energy label and energy rating system were revised to better differentiate the more energy-efficient models in the market.

Minimum Energy Performance Standards (MEPS): Introduced in 2011 to raise the average energy efficiency of products in the market.

Only household refrigerators, air-conditioners, motors, clothes dryers and lamps that meet the minimum energy efficiency standards can be sold in Singapore.

Average energy efficiency of air-conditioners and refrigerators have improved by about 23% and 39% respectively.

Water Efficiency Labelling Scheme (WELS) and Minimum Water Efficiency Standards: Introduced WELS, where suppliers were required to label the water efficiency of their water fittings and appliances.

Mandatory WELS covers taps and mixers, dual-flush low capacity flushing cisterns, urinal flush valves and waterless urinals, and washing machines.

Minimum water efficiency standards were introduced to phase out the least water - efficient products. For example, all taps, dishwashers and mixers sold or supplied in Singapore are required to meet at least “2-tick” (>5 to 7, >2 to 4, >4 to 6 litre/min flow rate for shower taps & mixtures, basin taps & mixers and sink/bib taps & mixers respectively, >3.5 to 4.0 (full flush), >2.5 to 3 (reduced flush) litres per flush for flushing cisterns, >0.5 to 1 litres per flush for urinal flush valve & waterless urinals and >3.5 to 4 litres per flush for WC flush valves, >9 to 12 litres per kg for domestic clothes washing machines and >0.9 to 1.2 litre per place setting for domestic dishwashers) water efficiency standard.

Water Efficiency Management Plan (WEMP): Introduced in 2010 as a voluntary initiative for commercial and industrial users to improve the efficiency of their water use.

All large water users who meet the water use threshold of 60,000 cubic metres must submit their WEMPs on an annual basis. These users also required to install water meters to measure and monitor water consumption.

WEMP includes an analysis of current water usage and proposed water conservation measures.

Water-Efficient Building: Water Efficient Buildings (Basic) certification programme encourages building owners to implement water efficiency measures.

WEB (Basic) certification obtained by installing water-efficient fittings and adopting water efficient flow rates/flush volumes.

Certified buildings typically save 5% of their monthly water consumption.

WEB (Basic) requirements are also recognised under the Green Mark Certification Scheme for buildings, which is an initiative to drive Singapore’s construction industry towards more environment-friendly buildings.

Water Efficiency Fund: Co-fund the implementation of water efficiency projects. Projects include feasibility studies, water audits, recycling efforts, use of alternate sources of water and community wide water conservation programmes.

Public Sector Taking the Lead: Public Sector Taking the Lead in Environmental Sustainability (PSTLES) initiative requires public agencies to implement measures for energy efficiency, water efficiency and recycling.

Requires each Ministry to appoint a Sustainability Manager, set sustainability targets, and develop a resource management plan.

Environmental sustainability efforts were outlined in the Public Sector Sustainability Plan 2017–2020.

Public agencies retrofitting major energy consuming equipment encouraged to adopt the Guaranteed Energy Savings Performance (GESP) contracting model. Under the model, an accredited energy services company conducts an energy audit of the facility, implements proposed energy efficiency improvement measures, and guarantees system performance and annual energy savings of the retrofitted equipment. As of February 2018, 32 large building owners called GESP contracts for building retrofit works. These building owners saved an average of 16% in electricity use, which is equivalent to annual savings of S\$11.3 million.

Challenges:

Future Waste Management: In 2017, Singapore generated about 7.7 million tonnes of waste.

Amount of waste generated is expected to increase in tandem with population and economic growth.

Household recycling rate was 21% in 2017.

Manpower constraints limit the resources available for our waste management industry.

Dealing with E-Waste: Improper disposal of e-waste could lead to environmental pollution and be detrimental towards human health and waste of precious natural resources

Currently, e-waste in Singapore is recycled on a voluntary basis by businesses and consumers.

Typically, only e-waste of high residual value is traded in or sold, while e-waste of lower residual value is disposed of along with general waste for incineration.

Opportunities:

Finding Innovative Solutions: Studying of benefits of smart waste collection systems to optimise waste collection operations, manpower and resources. This includes leveraging technologies such as bin fill sensors, a smart card access system for waste disposal chutes, and the use of side-loader bins that require only one operator to carry out recyclables collection.

Reaping synergies from the water-energy waste nexus at Integrated Waste Management Facility (IWMF), which will be integrated with the Tuas Water Reclamation Plant (TWRP) to allow for synergies such as effluent water from wastewater treatment being used for cooling waste incineration equipment; while food waste can be co-digested with used water sludge to enhance biogas production to increase the overall plant thermal efficiency. Integrating the facilities will also reduce carbon emissions by more than 200,000 tonnes annually.

E-Waste Management System: Mandatory e-waste management system will be implemented in 2021 to ensure the proper recycling of e-waste, where safety and environmental standards are adhered to for the public to have access to convenient avenues to recycle e-waste.

System would be based on the EPR concept and entail the assignment of responsibilities to key stakeholders in the e-waste value chain.

Love Your Food @ Schools: Two-year project launched in April 2017 to encourage youth to cherish food and reduce food wastage.

Closed-loop food waste management system was introduced in 10 participating schools to encourage students and staff to reduce generation of food waste, and to segregate and treat food waste using on-site food waste digesters to produce compost.

Students host neighbouring schools and community partners on learning journeys to spread awareness about the importance of food waste minimisation and recycling.

Compost generated is distributed during these learning journeys and to community gardens.

Project complements Singapore's holistic food waste management strategy in working towards becoming a Zero Waste Nation.

Semakau Landfill: Singapore has one landfill, located about eight kilometres south of mainland Singapore on the man-made island of Pulau Semakau.

Perimeter bund of the landfill is lined with an impermeable membrane and a layer of marine clay.

Thriving flora and fauna on and around Semakau Landfill indicative that the ecosystem has been well protected.

During the Phase II development of Semakau Landfill, over 700 colonies of corals were transplanted to Sisters Island Marine Park and more than 450 fishes were caught and transferred to open sea.

Open for educational visits.

Singapore is keenly aware that we cannot continually build landfills.

At the current rate at which we are depositing waste to Semakau Landfill, it will run out of space by 2035.

Goal 13: Climate Action

Measures:

Pursing Green Growth by Reducing Carbon Intensity: Aim to reduce Emissions Intensity by 36% from 2005 levels by 2030, and stabilise emissions with the aim of peaking around 2030.

Reduced greenhouse gas emissions by 16% from the business-as-usual level from 2009-2020.

Singapore is amongst the 15 best performing countries in terms of emissions intensity.

From 2000 to 2012, our greenhouse gas emissions grew by 2% per year, compared to a GDP growth of 5.7% per year over the same period.

Set up the Inter-Ministerial Committee on Climate Change (IMCCC) in 2007 to enhance Whole-of-Government coordination on climate change policies.

IMCCC headed by Deputy Prime Minister and Coordinating Minister for National Security and comprises Ministers from seven key agencies.

National Climate Change Secretariat (NCCS) was set up in July 2010 under the Prime Minister's Office to support the IMCCC.

Publicised our national strategy on climate change through several publications that have been released and revised iteratively, including National Climate Change Strategy (2012), Singapore Sustainable Blueprint (last updated in 2015), and two-pronged Climate Action Plan (2016): Take Action Today, for a Carbon-Efficient Singapore, and A Climate-Resilient Singapore, for a Sustainable Future.

Designated 2018 the Year of Climate Action in Singapore (YOCA). YOCA is aimed at raising the level of national consciousness on the need to take collective and individual action against climate change.

Key mitigation strategies outlined in Climate Action Plan: improving energy efficiency, reducing carbon emissions from power generation, developing and deploying cutting-edge low-carbon technologies and encouraging collective action among government agencies, individuals, businesses, and the community.

Public sector has adopted sustainability measures under Public Sector Taking the Lead in Environmental Sustainability (PSTLES) programme.

Under the Public Sector Sustainability Plan 2017-2020, the public sector achieved an electricity savings of 15%, water savings of 5% and 100% green building adoption.

Transport: Singapore's goal is for 75% of morning and evening peak journeys by 2030, and 85% by 2050, to be on public transport, up from 67% in 2017.

Vehicle Emissions Scheme (VES) provides incentives for the purchase of less pollutive vehicles and imposes surcharges on pollutive ones.

Singapore's first electric car-sharing programme, BlueSG, was launched in December 2017, with an aim to provide 1,000 shared electric vehicles and 2,000 charging points island-wide by 2020.

Buildings: Singapore aims to green 80% of all buildings by 2030, up from 31% in 2016. In line with our vision 'Positive-energy Low-rise, Zero-energy Medium-rise, Super Low-energy High-rise buildings in the tropics.'

Enhancement of the Building and Construction Authority of Singapore (BCA) Green Mark Scheme (a benchmarking scheme that incorporates internationally recognised best practices in environmental design and performance. Buildings, office interiors and parks can be certified under this scheme) and encouraging retrofitting of tenanted spaces.

Building Retrofit Energy Efficiency Financing by BCA provides loan-loss guarantees on loans made by financial institutions for energy efficiency retrofits.

Energy: 95% of electricity is generated today from natural gas.

Increase of solar photovoltaic deployment to 350 megawatt-peak (MWp) in 2020, and to 1 gigawatt-peak (GWp) beyond 2020.

Carbon Tax: First in Southeast Asia to introduce a price on carbon.

Carbon tax of S\$5 per tonne of carbon dioxide-equivalent emissions, in the first instance from 2019 to 2023. Applies to all sectors without exemption, covering about 80% of emissions. Intend to increase it to a rate of between S\$10 and S\$15 per tonne of emissions by 2030.

Strengthen Resilience, Adaptive and Institutional Capacity: Developed a Resilience Framework to take into consideration the latest science and changing needs to guide adaptation planning. Used to identify risks in six key areas for Singapore: coastal protection, water resources and drainage, biodiversity and greenery, public health and food security, network infrastructure, and building structures and infrastructure.

Established the Centre for Climate Research Singapore (CCRS) to develop in-house research expertise in the weather and climate of Singapore and the wider Southeast Asia region. In 2015, CCRS published the Second National Climate Change Study, which outlines long-term climate change projections for Singapore up to 2100.

BCA has commissioned a Coastal Adaptation Study to develop a national framework for Singapore's long-term coastal protection needs.

Study of how built environment and urban greenery could affect microclimatic conditions and identifying recommendations for the planning and design of public spaces and buildings.

Research institutes based in Singapore are conducting further research on climate change-related areas such as corals, algae blooms and tropical diseases, working with government agencies and collaborating with overseas counterparts.

Welcomed the relocation of the World Meteorological Organization (WMO)'s Regional Office for Asia and South-West Pacific to Singapore. The Regional Office, which is co-located with the Centre for Climate Research Singapore (CCRS), conducts programmes to enhance the region's understanding of climate science, and tackle challenges arising from climate change and extreme weather.

Promoting Climate Change Public Awareness and Action: As part of YOCA, an online website ClimateAction.sg was launched for individuals, corporates, NGOs and institutions to pledge their commitments and actions.

Created a special Climate Action SG Grant to provide funds for local constituencies, NGOs and interest groups to organise ground-up events and activities that increase awareness on climate action.

Climate Action SG Alliance, led by 3P partners, has also been set up to advocate climate action in Singapore through harnessing the power of public communications to raise awareness on climate issues and to translate this awareness into action.

Launched the revamped Sustainable Singapore Gallery at the Marina Barrage to raise awareness of Singapore's sustainability journey, and to remind Singaporeans of the challenges ahead for a sustainable Singapore.

International Partnerships: Shares experiences and best practices on climate change under the Singapore Cooperative Programme (SCP), in key areas such as sustainable development, urban planning, water and transport management.

Under the SCP, established the Sustainable Development and Climate Change (SDCC) programme in 2012. SDCC is tailored to meet the needs of developing country partners, including Small Island Developing States (SIDS) and the Least Developed Countries (LDCs). Trained around 3,600 officials from developing countries, including 1,600 from SIDS and LDCs.

Collaborated with the United Nations Office for Disaster Risk Reduction (UNISDR) to conduct two joint specialised training courses for 17 countries. Training courses provided practical support and knowledge for the implementation of the Sendai Framework for Disaster Risk Reduction.

Collaborated with the Japan International Cooperation Agency (JICA) to enhance the disaster management capabilities of other countries through the Disaster Risk Reduction and Response Course. Since its inception in 2015, more than 50 Governmental officials from 13 countries have been trained.

Actively supports and participates in the efforts led by the International Civil Aviation Organization (ICAO) and the International Maritime Organization (IMO) to address global emissions from international aviation and maritime transport.

At ICAO, Singapore contributes to the development of a holistic strategy and basket of measures to mitigate international aviation emissions, including the Carbon Offsetting and Reduction Scheme for International Aviation (CORSIA). Implemented

both mandatory and voluntary measures under the ICAO and committed to participate under CORSIA.

At IMO, contributed to the work of establishing energy efficiency measures for international shipping and to formulating a comprehensive strategy to reduce international maritime transport emissions.

Challenges:

Limited Scope for Deploying Renewable Energy: With limited land and geography, available resources, technology options, and other domestic considerations, faced with constraints on ability to deploy alternative energy.

Small size and high urban density limits the extent to which large amounts of alternative energy can be commercially deployed.

While solar energy is the most promising renewable energy option, competing land uses and high cloud cover means Singapore is not able to generate sufficient baseload electricity from solar based on current technologies.

Climate Science Expertise: Need to undertake efforts to enhance climate science capability and better understand the diverse impacts of climate change on Singapore and the wider region.

Opportunities:

Develop Options to Further Deploy Renewable Energy: To increase solar photovoltaic (PV) deployment in Singapore to 350 MWp by 2020, and to 1 GWp beyond 2020, Singapore will be investing in R&D as well as test-bedding to improve the performance of solar PV systems and develop innovative ways of integrating solar and other alternative energy systems into urban environment.

Projects include: HDB launched initiative to install solar panels on rooftops of high-rise apartments.

Economic Development Board's (EDB) and PUB's floating photovoltaic PV project currently pilots ten systems of 1 MWp total floating solar panel installations on water surfaces at Tengeh Reservoir.

Singapore's first long-span wind turbine was installed at an offshore landfill in October 2017. It is sensitive enough to generate power with wind speeds as low as 3 metres per second. Potential to develop "hybrid micro-grids" in the next few years.

SolarNova programme led by EDB and HDB aggregates solar demand across government agencies to build up local expertise in solar manufacturing, project development, system integration, financing, and to encourage greater adoption of solar energy.

Further Enhance Resilience: Need to continue to collaborate and share information on climate science and challenges arising from climate change impacts with international partners through various channels, including the WMO Regional Office for Asia and the South-West Pacific located in Singapore.

Exploring opportunities with the WMO Regional Office to enhance cooperation among the national meteorological and hydrological services in the 58 states and territories under its charge, and with the wider scientific community.

South-eastern Asia-Oceania Flash Flood Guidance (SAOFFG) aims to provide real-time information on small-scale flash floods to disaster management agencies in Brunei Darussalam, Indonesia, Malaysia, Papua New Guinea, the Philippines, Singapore and Timor-Leste.

Coastal Adaptation and Resilience: Singapore's daily mean temperature is projected to rise by up to 4.6 degrees Celsius towards the end of the century, with more than 300 warm days a year compared to an average of 30 today.

Much of Singapore lies within 15m above the mean sea level, and 30% of Singapore's ground level is less than five metres above mean sea level.

Mean sea level is estimated to rise by up to 0.76 metres in the long term.

In 2011, minimum reclamation level for new projects was raised from three metres to four metres above mean sea level, to ensure that the new reclaimed land is safeguarded against sea level rise.

Raised the minimum crest levels for entrances to underground facilities such as MRT stations.

At Changi Airport, the future Terminal 5 will be built 5.5 metres above the mean sea level as a precaution.

Significant upgrades to the drainage system are in progress, with the installation of detention tanks and pumps.

Cooling Singapore Microclimate Research Project: In 2017, Singapore experienced the warmest year on record (27.7C) that was not influenced by an El Nino event. In 2016 and 2015, annual average temperatures were at record highs of 28.4 and 28.3 degrees Celsius.

High humidity inhibits cooling by evaporation of perspiration.

Singapore suffers from the Urban Heat Island (UHI) effect which leads to higher temperatures, as urban materials and surfaces trap and radiate more heat.

Multi-disciplinary research team led by the Singapore-ETH Centre, in collaboration with the National Research Foundation. Singapore MIT Alliance for Research and Technology (SMART), Technical University of Munich (TUM CREATE), and the National University of Singapore exploring the means to mitigate the UHI effect in Singapore.

Goal 14: Life below Water

Measures:

Integrated Urban Coastal Management: Adapted the Partnership in Environment Management for the Seas of East Asia (PEMSEA) Integrated Coastal Management principles to our unique context by incorporating an urban perspective into the framework.

Integrated Urban Coastal Management strategy has four guiding principles: Proactive planning and management to safeguard our coastal and marine environment by optimising the use of coastal resources, including coastal spaces in a sustainable manner, a Whole-of-Government approach to ensure consultative planning and coordination of policies between all stakeholders in coastal and marine land use and planning, active partnerships through community engagement & public awareness programmes; and science-based management through research, monitoring, restoration and enhancement programmes to conserve sensitive coastal habitats and their biodiversity and natural resources amidst coastal development.

A Marine Park for All: Sisters' Island Marine Park (SIMP) was established in 2015 as Singapore's first marine park. Located to the south of mainland Singapore on Sisters' Island, SIMP spans 40 hectares, and its surrounding areas and houses a wide range of marine habitats, including coral reefs, sandy shores and seagrass areas. SIMP showcases subtidal coral reef and shallow sea-floor areas, in addition to the unique biodiversity within the inter-tidal areas and coastal forests. IMP provides Singaporeans and visitors a unique recreational experience while educating them about marine natural heritage. At the same time, SIMP protect and safeguard our rich marine biodiversity, facilitates cutting-edge research, and provides opportunities to test-bed habitat rehabilitation, restoration and enhancement technologies.

Marine Conservation Programmes: Singapore's marine conservation and management strategies captured under the Marine Conservation Action Plan (MCAP), which is an evolving plan grounded in science to meet the current and future conservation needs of Singapore's coastal and marine environment.

Initiated the following marine species recovery programmes: Marine turtles (*Eretmochelys imbricata* and *Chelonia mydas*), giant clams (*Tridacna squamosa*, *T. maxima* and *T. gigas*), Neptune's Cup Sponge (*Cliona patera*) and several locally rare hard and soft coral species.

SIMP is the focal point for the species recovery programme, where strategies and methods are tested and monitored before applying them to other coastal and marine areas in Singapore.

Installed biodiversity enhancement units around Singapore to help existing biodiversity within Singapore's coastal and marine habitats flourish. These include the creation of inter-tidal pools and surface complexity enhancements along coastal seawalls, and the use of enhanced floating structures that increase visibility and bring marine biodiversity closer to the people.

Involving the Community as Stakeholders: Building on the pool of volunteers from community engagement projects like the Comprehensive Marine Biodiversity Survey and MCAP keeps these volunteers engaged in nature appreciation and conservation through talks and workshops, SCUBA diving, citizen science activities and nature ambassador opportunities.

SIMP and other initiatives like the Ubin Living Lab are focal points for complementary community stewardship programmes under the MCAP. Activities such as guided inter-tidal walks have begun at the SIMP, as well as other outreach activities and events organised in conjunction with a number of NGO stakeholders such as the Marine Conservation Group of the Nature Society of Singapore, Our Singapore Reefs, Hantu Bloggers and various academic partners.

Maritime and Port Authority of Singapore (MPA) also organises outreach programmes to raise awareness and inspire youths, schools, and members of the public to play a part in protecting the marine environment.

Research and Test-Bedding of New Technologies: Invested in key areas of applied research that will help fill knowledge gaps in the development and implementation of nature conservation policies.

Research at the SIMP addresses marine genetic connectivity, climate change and its impacts on marine biodiversity, and the application of clean and renewable sources of energy.

MPA and the National University of Singapore's Centre for International Law (CIL) established the CIL-MPA Oceans Governance Research Programme in April 2016 to contribute to the greater understanding of maritime law and ocean governance.

Sustainable Maritime Transport: Maritime Singapore Green Initiative (MSGI) was launched in 2011 to reduce the environmental impact of shipping and shipping-related activities on the coastal and marine environment.

MPA pledged to invest up to S\$100 million over five years under the MSGI's five programmes: the Green Ship Programme (Qualifying ships that adopt energy efficient ship designs exceeding IMO's Energy Efficiency Design Index and/or adopt approved SOx scrubber technology exceeding International Maritime Organisation (IMO)'s emission requirements can enjoy up to 75% reduction of Initial Registration Fees and up to 50% rebate on Annual Tonnage Tax.), Green Port Programme (Ships that use approved abatement technology or burn clean fuel while at berth can enjoy 15% reduction in port dues, while ships that do so throughout their entire port stay within the port limits can enjoy 25% reduction) and Green Technology Programme (companies can apply for grants of up to 50% of qualifying costs to co-fund the development and adoption of green technological solutions).

MPA has in place the Marine Emergency Action Procedure to effectively manage oil and chemical spills, as well as other marine incidents.

Multi-agency ChemSpill exercise simulates a chemical spill and tests agencies' capabilities and co-operation to combat chemical pollution. The exercise is conducted every alternate year in conjunction with the biennial International Chemical and Oil Pollution Conference and Exhibition (ICOPCE).

In 2015 and 2017, worked with the International Maritime Organization (IMO) to co-organise two editions of the Future-Ready Shipping (FRS) Conference in Singapore. FRS Conference gathered maritime leaders and professionals worldwide to exchange views and foster collaboration on energy-efficient maritime transport technologies. Also served as a platform for capacity building for developing countries and Small Island Developing States.

Worked with the IMO to provide capacity building courses for countries under the MPA-IMO Third Country Training Programme (TCTP). MPA-IMO TCTP offers courses on the implementation of relevant IMO instruments, including those that protect the marine environment through the prevention of pollution from ships.

Challenges:

Habitat Fragmentation: Habitats are fragmented and occur in isolated patches.

Habitat and Biodiversity Loss: Due to rapid pace of Singapore's growth and limited land, shallow coastal and marine areas can be lost with urbanisation.

Opportunities:

Maintaining Ecological Stepping Stones: Unlike fragmented terrestrial habitats that can become ecologically isolated, fragmented but hydro-dynamically connected coastal and marine areas can generally maintain ecological connectivity through a network of "stepping stones" – smaller habitat patches or fragments that facilitate movement of populations between larger ones. A study of the ecological connectivity of Singapore's mangrove, inter-tidal and sub-tidal habitats indicated that despite the spatial fragmentation, ecological connectivity continues to be maintained across all habitats due to the continued presence of these "stepping stones".

A key strategy in Singapore's coastal and marine conservation efforts is to map out these "stepping stones" to ensure they are adequately maintained for continued ecological connectivity.

Habitat Enhancement, Restoration and Species Recovery: Coastal protection and mangrove restoration biodiversity project was implemented along a severely degraded mangrove area which incorporated both hard and soft engineering solutions to arrest coastal erosion and restore the mangroves.

Purpose-built and nature inspired inter-tidal units were fabricated and installed along one of Singapore's armoured coastal revetment to increase surface complexity and provide micro-niches to enhance inter-tidal biodiversity.

Habitat enhancement and restoration framework was developed for the SIMP, with the implementation of reef enhancement units to enhance degraded reef areas coupled with a donor-supported Plant-a-Coral, Seed-a-Reef species recovery programme to propagate and out-plant locally rare coral species.

Giant clam propagation and reintroduction programme initiated to reintroduce the locally extinct *Tridacna gigas* species and increase the numbers of two other locally rare *T. squamosa* and *T. maxima* species.

Neptune Cup Sponge: Population of the Neptune's Cup sponge declined rapidly and it was last sighted in the 1870s. Decline appeared to have taken place elsewhere as well, with a live specimen last collected off Bantam in West Java, Indonesia in 1908, leading many scientists to believe that the Neptune's Cup sponge had become globally extinct.

In March 2011, marine biologists from a Singapore-based company discovered a unique-looking sponge, which turned out to be a young Neptune's Cup sponge. Another specimen was soon found within a 50 metre radius of the first. As of 2018, five Neptune's Cup sponges have been found in Singapore waters.

Relocated the five sponges to the Sisters' Island Marine Park where scientists are currently studying the specimens, including how they reproduce. End goal is to rebuild the population of the Neptune's Cup sponge so that it never goes extinct.

Goal #15: Life on Land

Measures:

Saving Species: Over 500 species were discovered or rediscovered locally by the National Parks Board (NParks), research partners and members of the public.

Identified 60 endemic, rare or threatened native species of plants and animals as conservation priorities, and plan to increase their populations in Singapore through reintroduction, habitat enhancement and protection. These target species were identified based on their conservation status and distribution under Singapore's Red Data Book (publication by the Nature Society (Singapore) on endangered plants and animals in Singapore).

Globally critically endangered Singapore freshwater crab *Johora singaporensis* only found in a few local hill streams in Singapore.

No wild species of *Marsdenia maingayi*, a plant, sighted after 1920s.

Bringing Back Habitats: Habitat Enhancement and Restoration Framework was adopted to standardise the approach in line with accepted science-based methods. Site assessments are first carried out to document and understand the health of habitats, and suitable restoration techniques are subsequently determined.

Steps may then be taken to enhance site conditions through soil remediation or improving water quality.

Peripheries of existing habitats affected by surrounding urban areas are enhanced.

To mitigate impact of edge effects that arise from fragmentation and safeguard the integrity of nature reserves, buffers in the form of nature parks widen recreational areas for a larger segment of users without severely impacting the core of nature reserves.

A Network of Pervasive Greenery: Four nature reserves, more than 350 parks and gardens, and over 300 kilometres of linear parks known as park connectors in Singapore.

Extensive roadside greenery which connects habitats, including strategic green corridors, also known as Nature Ways, along roadsides that connect areas of high biodiversity.

Trees, shrubs and ground cover planted to mimic the emergent, mid-canopy, understorey and undergrowth layers of a natural forest in these Nature Ways to link fragmented natural habitats and enhances biodiversity in urban environment.

Skyrise greenery, in the form of roof gardens and vertical green walls, augment ground level greenery and provide refuge to birds, bats and insects within the urban landscape.

Eco-Link@BKE, an hourglass-shaped ecological bridge across the Bukit Timah Expressway, connects Bukit Timah and Central Catchment Nature Reserves which were once together to mitigate fragmentation effects due to the construction of the expressway in 2011. Native plants were planted on the bridge to encourage its use by the fauna on either side and monitoring programmes are in place to ascertain the effectiveness of the bridge.

Involving the Community as Stakeholders: Party to the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES).

Measuring our Efforts: Initiated the development of an index in 2008 designed to benchmark conservation efforts at the city-level, the Singapore Index on Cities' Biodiversity (SI).

Worked with the Convention on Biological Diversity (CBD) Secretariat and the Global Partnership on Sub-National Governments and Cities for Biodiversity to develop the SI, together with an international Technical Task Force.

SI is intended to be used as a self-assessment tool to monitor cities' progress in biodiversity conservation efforts over time.

Comprises 23 indicators that measure native biodiversity, ecosystem services provided by biodiversity, and governance and management of biodiversity.

Several cities worldwide adopted the SI, such as Auckland, Helsinki, Nagoya, Hyderabad and Bangkok, with most finding that the process has facilitated capacity building in biodiversity conservation by providing biodiversity conservation guidelines and in setting priorities for conservation actions and budget allocation.

Challenges:

Habitat Fragmentation: Flora and fauna require a minimum area to sustain their populations.

Edges of habitats are more susceptible to fragmentation, i.e. edge effects, which are largely attributed to air, noise and light pollution from developed areas.

Habitat Loss: Need to restore habitats that have been lost and enhance existing habitats.

Opportunities:

Establishing Connectivity Between Natural Areas: Extensive park connector network with a total length of 306 kilometres in 2017.

Roadside greenery encourages connectivity in the form of nature ways, which emulate the structural complexity of a natural forest using mixed species planting.

Small patches of greenery on rooftops and incorporated into developments act as stepping stones between natural areas.

Aim to have 400 kilometres of park connectors, 180 kilometres of Nature Ways and 200 hectares of skyrise greenery by 2030.

Bishan-Ang Mo Kio Park Habitat Restoration: In 2009, reopened Bishan-Ang Mo Kio Park, located in central Singapore, after habitat restoration and enhancement and design upgrade.

Park originally contained a concrete canal that ran its entire length. Redeveloped under the Active, Beautiful, Clean (ABC) Waters Programme into a meandering naturalised river known as Kallang River.

Through collaboration between PUB and NParks, Kallang River was integrated into the park's surroundings through the use of natural materials to shore up its river bank. These efforts attracted wildlife.

ABC Waters design comprises plants and planting media which have also been incorporated into the park to maintain the water quality in the ponds and the river naturally, without the use of chemicals. For instance, cleansing biotopes that filter and absorb pollutants are located upstream. Water is also recycled for use in the water playground after undergoing UV treatment. The landscape surrounding the river is also designed to accommodate increases in water levels during heavy rainfall.

Skyrise Greenery in Singapore: Urban Redevelopment Authority and NParks introduced the Landscaping for Urban Spaces and High-rises (LUSH) programme and Skyrise Greenery Incentive Scheme. These offer building owners and developers incentives such as additional gross floor area for rooftop greenery installation and co-funding for skyrise greenery installation.

Recognised creative skyrise greenery designs through the Skyrise Greenery Awards. In 2017, received a record 177 entries for the Awards. Past award winners include Kampung Admiralty (2017), an integrated development combining residential, commercial and community facilities, PARKROYAL on Pickering (2013) and the Khoo Teck Puat Hospital (2010), where the incorporation of greenery creates a healing environment to boost patient recovery.

As of 2017, there were 100 hectares of skyrise greenery in Singapore. This has helped to increase the green coverage, as evidenced by Singapore's second place out of 21 cities ranked in the Green View Index (index by the Massachusetts Institute of Technology's Senseable City Lab, measures the canopy cover in cities. The Index is calculated using Google Street View panoramas to proxy human perception of the environment from the street level).

Intent to double the amount of skyrise greenery in Singapore to 200 hectares by 2030 under the Sustainable Singapore Blueprint.

Goal 16: Peace and Justice strong Institutions:

Measures:

Equal Protection According to Men, Women and Children: Ranked first in Gallup's Law and Order Report from 2014 to 2017 (report for countries to compare safety to other countries).

Party to the Convention on the Elimination of All Forms of Discrimination against Women (CEDAW).

Singapore Police Force works closely with the National Crime Prevention Council, an NGO, to raise awareness on sexual offences.

Children's and Young Persons Act (CYPA) safeguards the care, protection and rehabilitation of children and young persons below 16 years of age. Amended in 2011 to improve the protection for children and young persons, such as exempting a child from attending court proceedings in cases where the child could be adversely affected and safeguarding a child's privacy by prohibiting the publication of personal information.

Protection from Harassment Act criminalises different forms of harassment such as sexual harassment and school or cyberspace bullying.

Eradication of Corruption: Ranked the 6th least corrupt country by Transparency International's Corruption Perceptions Index 2017, and the least corrupt country in the Political and Economic Risk Consultancy's 2017 Report on Corruption in Asia.

Established two legislations: Prevention of Corruption Act (PCA) and the Corruption, Drug Trafficking and Other Serious Crimes Act (CDSA). PCA applies to those who give or receive bribes in both the public and private sector. The CDSA, when invoked, confiscates ill-gotten gains from corrupt offenders.

Established an independent judiciary free from political interference.

Public service is guided by a strict Code of Conduct which sets out the high standards of behaviour expected of public officers based on integrity, incorruptibility, and transparency.

Anti-corruption is rigorously enforced by the Corrupt Practices Investigation Bureau (CPIB), which is an independent national anti-corruption agency responsible for investigating and preventing corruption in Singapore.

Effective, Accountable and Transparent Institutions: Singapore Constitution lays down the fundamental principles and basic framework for the separation of powers between the Executive, the Legislature, and the Judiciary to guarantee a system of checks and balances. The Constitution also guarantees fundamental liberties such as the equal protection of all persons before the law.

Judiciary is the guardian of the Constitution. Singapore's Supreme Court is empowered to review laws and Executive actions to ensure that they are constitutional.

92% of respondents expressed confidence in our legal system. Over 90% believed that the administration of justice by State Courts (i.e. the lower Courts) was carried out with integrity and in a fair and independent manner.

Ranked 19th by the World Economic Forum for judicial independence, and 2nd by the Heritage Foundation's 2017 Index of Economic Freedom for judicial effectiveness.

Pro bono (professionals who do not take payment of services) legal services are provided through a partnership between the Government, the Law Society of Singapore, various volunteers and volunteer welfare organisations.

For low-income individuals facing legal issues, the Legal Aid Bureau provides civil legal aid for most civil proceedings, including judicial reviews, while the Law Society Pro Bono Services runs the Criminal Legal Aid Scheme (CLAS).

Number of CLAS applicants who received full representation or unbundled legal services increased fourfold between 2014 and 2017.

Strength of Singapore's public institutions was ranked 2nd by the World Economic Forum's Global Competitiveness Report (2017-2018).

Singapore Public Service employs 145,000 officers in 16 Ministries and more than 50 Statutory Boards. Work of these agencies is broadly characterised into five sectors: economic, security, social, infrastructure and environment, and central administration.

Whole-of-Government (WOG) approach, which calls for the rigorous coordination between and within agencies on the planning, creation, and execution of public policies for maximum effectiveness.

Civil Service College (CSC) established as the main training arm of the Singapore Public Service. CSC offers 468 training programmes on a broad range of topics such as Data Analytics, International Relations, and Public Finance for public servants of different ranks.

Parliament of Singapore is central to the transparency and accountability of Government. Members of Parliament actively raise the people's concerns, debate policies, and pose questions to the Executive to seek explanations for the public's understanding and benefit.

Government's main Feedback Unit, REACH (Reaching Everyone for Active Citizenry @ Home) gather and gauge public ground sentiments on issues of concern, reach out and engage Singaporeans through various media platforms, and promote active citizenry by encouraging citizen participation in shaping Singapore's policies.

Specialised Councils under the Presidency have also been established to ensure that the voices of minority groups are taken into consideration in policymaking. For example, the Presidential Council for Minority Rights is tasked with examining legislation to ensure that they are not disadvantageous to any racial or religious community in Singapore.

Low Crime in Singapore: In the 2017 national Public Perception Survey, 92% of respondents rated general safety and security in Singapore favourably and 93% of respondents felt safe walking in their neighbourhood at night.

Stemming the Flow of Illicit Funds and Arms: Between 2008 and 2014, 66% of Singapore's money-laundering investigations and 27% of money-laundering convictions relate to criminal offences committed overseas.

Financial Action Task Force (FATF) and Asia/Pacific Group on Money Laundering (APG) Mutual Evaluation on Singapore assessed that Singapore possessed a strong

legal and institutional framework for combatting money laundering, terrorism financing, and proliferation financing.

Illegal manufacture and unauthorised trading of Small Arms and Light Weapons (SALW) are offences under Arms and Explosives Act. Strict processes must be observed for the legal manufacture of all SALW and exports to countries under the relevant sanctions imposed by the United Nations Security Council are not authorised.

Trafficking in Persons: Inter-Agency Taskforce was established in 2010 to ensure WOG coordination on TIP issues.

Taskforce launched the National Plan of Action (NPA) 2012-2015, which centres on a “4Ps strategy” of Prevention, Prosecution, Protection and Partnership.

Taskforce worked with the relevant stakeholders to develop a new National Approach against TIP, which was launched on 10 March 2016.

Enacted the Prevention of Human Trafficking Act.

Acceded to the United Nations Protocol to Prevent, Suppress and Punish Trafficking in Persons, especially Women and Children (UN TIP Protocol) in September 2015.

In January 2016, ratified the ASEAN Convention against Trafficking in Persons, Especially Women and Children (ACTIP).

Challenges:

Complex Future Operating Landscapes: Future operating landscape will be increasingly complex, with more people-to-people flows across borders, new threats, and disruptive technologies.

Relevance of Legislation: Need for legislation to be updated and enhanced to deal with new challenges in changing environment.

Opportunities:

Disruption with Technology: In crisis situations, UAVs can be deployed to provide a bird's-eye view of the ground situation and allow officers to make better informed decisions.

introduction of self-service booths means no need to deploy personnel to man immigration counters so that more manpower can be redeployed to security functions, such as monitoring suspicious travellers and managing incidents on the ground.

Review and Update of Legislation: Offers opportunities to review and update existing legislation to consider evolving circumstances.

Protecting the Youth under the UN Convention on the Rights of the Child: Commissioned a series of children's storybooks to educate children and their caregivers on children's rights in a light-hearted and reader-friendly manner.

Goal 17: Partnerships to achieve Sustainable Development

Measures:

The Singapore Cooperation Programme: Technical Assistance Programme

Over 119,000 officials from more than 170 countries joined SCP programmes in areas such as education, transport, economic development and trade promotion, healthcare, judiciary and public administration.

Supporting the 2030 Agenda: the Sustainable Development Programme: Launched the Sustainable Development Programme (SDP) under the SCP in 2015.

SDP aims to support developing countries' achievement of the SDGs by building capacity at three levels – leadership, city, and community – through partnerships with UN agencies and local NGOs.

SDP is specifically tailored for developing countries, in particular, the Small Island Developing States (SIDS) and the Least Developed Countries (LDCs), with a variety of courses on SDG-specific areas.

At the leadership level, partnered the UN Development Programme (UNDP) Global Centre for Public Service Excellence (GCPSE) to conduct “Transformational Leadership for the 2030 Agenda (TL2030)” programme, an invitation-only leadership course focusing on key challenges faced by Public Service leaders in fulfilling their national development aspirations in achieving the SDGs. Both sides also jointly organised the “Public Service: 2030 And Beyond (PS2030)” programme, targeted at senior-level officials involved in strategic policymaking and driving public sector reform towards implementing the SDGs to share insights and experiences on SDG implementation.

At the city level, Singapore has partnered UN-Habitat (organisation that supports countries to improve living conditions for all through sustainable urbanization) on a capacity building programme in support of the UN's New Urban Agenda and the achievement of SDG 11 on sustainable cities. Emphasis is to demonstrate how local development challenges can be overcome through applying sustainable urban system principles and long-term integrated master planning and development.

At the community level, Singapore is partnering NGOs to help build capacity on water, sanitation and hygiene issues in rural areas across Southeast Asia.

Supported Lien Aid and Mercy Relief to provide sustainable water solutions for around 21,000 people.

Partnered the UN International Children's Emergency Fund (UNICEF) to provide capacity building on strategies and tools on water, sanitation and hygiene services for Southeast Asian officials.

Third Country Training Programme: Third Country Training Programme (TCTP) framework of the SCP provides technical assistance in collaboration with other countries and international organisations.

Partnered more than 40 countries and international organisations, such as the United Nations Framework Convention on Climate Change (UNFCCC), UNDP and UN Environment.

Worked with Australia and the UNDP/UN Environment Global Support Programme to host annual regional capacity building workshops on transparency under the UNFCCC to assist countries with their national climate plans and strengthening their institutional arrangements, in order to implement their obligations under the UNFCCC.

To facilitate North-South cooperation, formed partnerships with other countries, such as the Singapore-US TCTP, the Japan-Singapore Partnership Programme for the 21st Century and the Australia-Singapore Partnership Arrangement for International Development.

Facilitate South-South cooperation with fellow developing countries, through collaborations with our neighbours from ASEAN as well as countries from outside the region to share unique and different experiences, such as the Singapore-Argentina TCTP, Singapore-Chile TCTP, and the Singapore-Mexico TCTP.

Singapore's ASEAN Chairmanship: Assumed the ASEAN Chairmanship in 2018 under the theme of "Resilience and Innovation". In line with the theme of the 2018 HLPF, "Transformation towards sustainable and resilient societies".

ASEAN Leaders' Vision Statement endorsed by the 32nd ASEAN Summit in April 2018 calls for ASEAN to become a sustainable community which promotes economic and social development alongside environmental protection, while leaving no one behind.

Leaders reaffirmed their commitment to the implementation of the 2030 Agenda, including through working with ASEAN's Dialogue Partners and external parties to promote complementarities between the ASEAN Community Vision 2025 and the 2030 Agenda through feasible and concrete projects.

ASEAN Smart Cities Network: Aims to facilitate cooperation on smart cities development, catalyse bankable projects within the private sector, and secure funding and support from external partners.

Core elements of the ASCN include city-specific Action Plans for Smart City Development, an ASEAN Smart Cities Framework, an Annual Meeting of the ASCN, and a Twinning Programme with cities from ASEAN's external partners.

First ASCN annual meeting convened alongside the World Cities Summit in Singapore in July 2018.

ASCN will officially launched by the ASEAN Leaders at the 33rd ASEAN Summit in Singapore in November 2018.

Special ASEAN Meeting on Climate Action: Intended to highlight ASEAN countries' commitment to collectively address climate change, including through the achievement of SDG 13 on climate action, and through fulfilling our commitments under the Paris Agreement. Discussions at SAMCA intended to feed into the Talanoa Dialogue 2018 at the 24th Conference of the Parties (COP) to the UNFCCC. The SAMCA also held back-to-back with an Expanded SAMCA involving the Presidents of the 23rd and 24th COPs to the UNFCCC, Fiji and Poland respectively, as well as ASEAN's external partners under the ASEAN Plus Three.

Multi-stakeholder Approach: Society Partners: Singapore supports a multi-faceted, multi-stakeholder approach to implementing the SDGs.

Engaging the youth: Launched a youth video competition, “The Sustainable Development Goals – What YOUth Can Do”. asked youth to describe an idea or specific solution to help achieve, in Singapore or around the world, one of the five SDGs that will be reviewed at the 2018 HLPF, or to depict the theme of the 2018 HLPF.

Novel and bright ideas on topics such as sustainable urban farming and the implementation of block-chain solutions to establish electrical infrastructure came from the answers.

Industry Efforts: Singapore’s Temasek Foundation, a Singapore-based non-profit philanthropic organisation, has been hosting annual “Ecosperity” Conferences in partnership with the Business and Sustainable Development Commission since 2014 to bring private and public sector leaders together to explore sustainable growth ideas.

Temasek teamed up with UNLEASH, a global non-profit innovation lab for solutions to the SDGs.

Hosted the second iteration of the UNLEASH annual event in May-June 2018, which brought together young, creative and innovative minds to harness the power of human ingenuity and create imaginative solutions that achieve the SDGs.

Businesses such as property developer City Developments Limited have launched sustainability blueprints that specifically consider the SDGs in their work.

Others, such as the Development Bank of Singapore, have also released green bonds as a form of sustainable financing, incentivised by the Monetary Authority of Singapore’s green bond grant scheme.

Worked closely with organisations like the Global Compact Network Singapore (GCNS), the local chapter of the UN Global Compact, to promote corporate sustainability efforts among companies in Singapore to encourage and facilitate efforts of companies which are committed to aligning their operations and strategies with sustainable development. Key activities include raising awareness of corporate social responsibility (CSR) initiatives among businesses through events and training programmes on topics such as “Sustainability Reporting” and “Climate Change and Business”, and nurturing the next generation of CSR leaders through youth initiatives.

Civil Society, Academia and Co-operative Organisations: Singapore National Co-operative Foundation (SNCF) has enacted measures in line with the SDGs in order to enhance their offerings. Includes ensuring price stabilisation of food supplies during crises, contributing to the provision of healthcare through its network of doctors, increasing women's access to economic opportunities, and facilitating access to education through providing awards, study grants and bursaries to explore how Singapore as a financial hub can offer green financing through a range of financial services and products, to promote the efficient flow of capital towards activities that are more sustainable and responsive to climate concerns.

Partnered Singaporean NGO, Lien AID, to establish two Community Water Enterprises in Cambodia to provide affordable treated drinking water to the local communities on a sustainable basis.

Actively engaged the World Toilet Organisation (WTO), a Singapore-based international non-profit organisation in its efforts to bring about better sanitation and toilet hygiene to developing countries such as India and China.

Challenges:

Encouraging SDG-oriented Mindsets: Raising the level of national consciousness for behavioural change, especially the need to take individual and collective action to achieve the SDGs is a challenge.

Opportunities:

Working with Youth: Anchor Green Primary School has partnered with IKEA to campaign for a sustainable living environment. Their projects include converting recycled fabric into bookmarks which are sold to help needy pupils.

Launched the Year of Climate Action in 2018 to encourage Singaporeans to take action to reduce our carbon footprint for a more sustainable future. Singaporeans are encouraged to make a Climate Action Pledge to verbalise actions they intend to take to reduce their individual carbon footprints.

